



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Lecture 6: Introduction to Probability

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Department of Management Studies



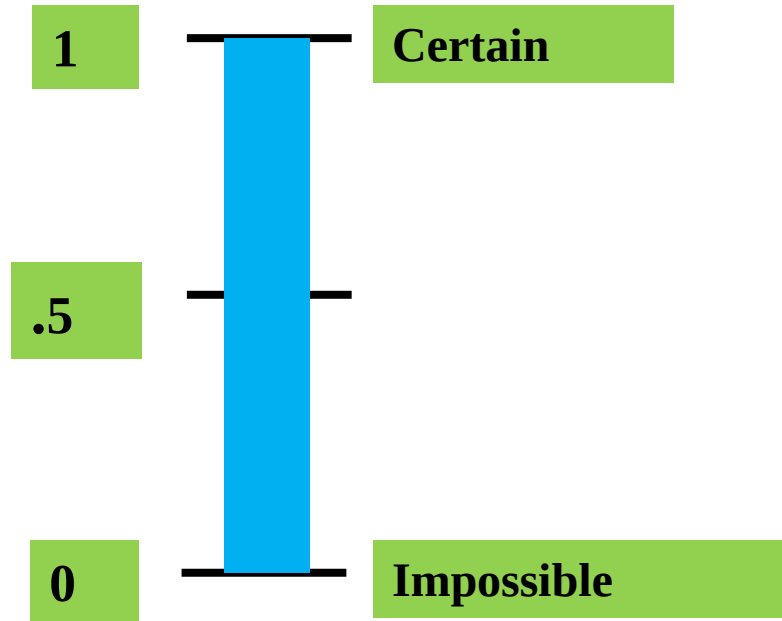
Lecture objectives

- Comprehend the different ways of assigning probability
- Understand and apply marginal, union, joint, and conditional probabilities
- Solve problems using the laws of probability including the laws of addition, multiplication and conditional probability
- Revise probabilities using Bayes' rule

Probability

- Probability is the numerical measure of the likelihood that an event will occur.
- The probability of any event must be between 0 and 1, inclusively
 - $0 \leq P(A) \leq 1$ for any event A.
- The sum of the probabilities of all mutually exclusive and collectively exhaustive events is 1.
 - $P(A) + P(B) + P(C) = 1$
 - A, B, and C are mutually exclusive and collectively exhaustive

Range of Probability



Methods of Assigning Probabilities

- Classical method of assigning probability (rules and laws)
- Relative frequency of occurrence (cumulated historical data)
- Subjective Probability (personal intuition or reasoning)

Classical Probability

- Number of outcomes leading to the event divided by the total number of outcomes possible
- Each outcome is equally likely
- Determined *a priori* -- before performing the experiment
- Applicable to games of chance
- Objective -- everyone correctly using the method assigns an identical probability

Classical Probability

$$P(E) = \frac{n_e}{N}$$

Where:

N = total number of outcomes

n_e = number of outcomes in E

Relative Frequency Probability

- Based on historical data
- Computed after performing the experiment
- Number of times an event occurred divided by the number of trials
- Objective -- everyone correctly using the method assigns an identical probability

Relative Frequency Probability

$$P(E) = \frac{n_e}{N}$$

Where:

N = total number of trials

n_e = number of outcomes
producing E

Subjective Probability

- Comes from a person's intuition or reasoning
- Subjective -- different individuals may (correctly) assign different numeric probabilities to the same event
- Degree of belief
- Useful for unique (single-trial) experiments
 - New product introduction
 - Initial public offering of common stock
 - Site selection decisions
 - Sporting events

Probability - Terminology

- Experiment
- Event
- Elementary Events
- Sample Space
- Unions and Intersections
- Mutually Exclusive Events
- Independent Events
- Collectively Exhaustive Events
- Complementary Events

Experiment, Trial, Elementary Event, Event

- **Experiment:** a process that produces outcomes
 - More than one possible outcome
 - Only one outcome per trial
- **Trial:** one repetition of the process
- **Elementary Event:** cannot be decomposed or broken down into other events
- **Event:** an outcome of an experiment
 - may be an elementary event, or
 - may be an aggregate of elementary events
 - usually represented by an uppercase letter, e.g., A, E1

An Example Experiment

- Experiment: randomly select, without replacement, two families from the residents of Tiny Town
- Elementary Event: the sample includes families A and C
- Event: each family in the sample has children in the household
- Event: the sample families own a total of four automobiles

Tiny Town Population		
Family	Children in Household	Number of Automobiles
A	Yes	3
B	Yes	2
C	No	1
D	Yes	2

Sample Space

- The set of all elementary events for an experiment
- Methods for describing a sample space
 - roster or listing
 - tree diagram
 - set builder notation
 - Venn diagram

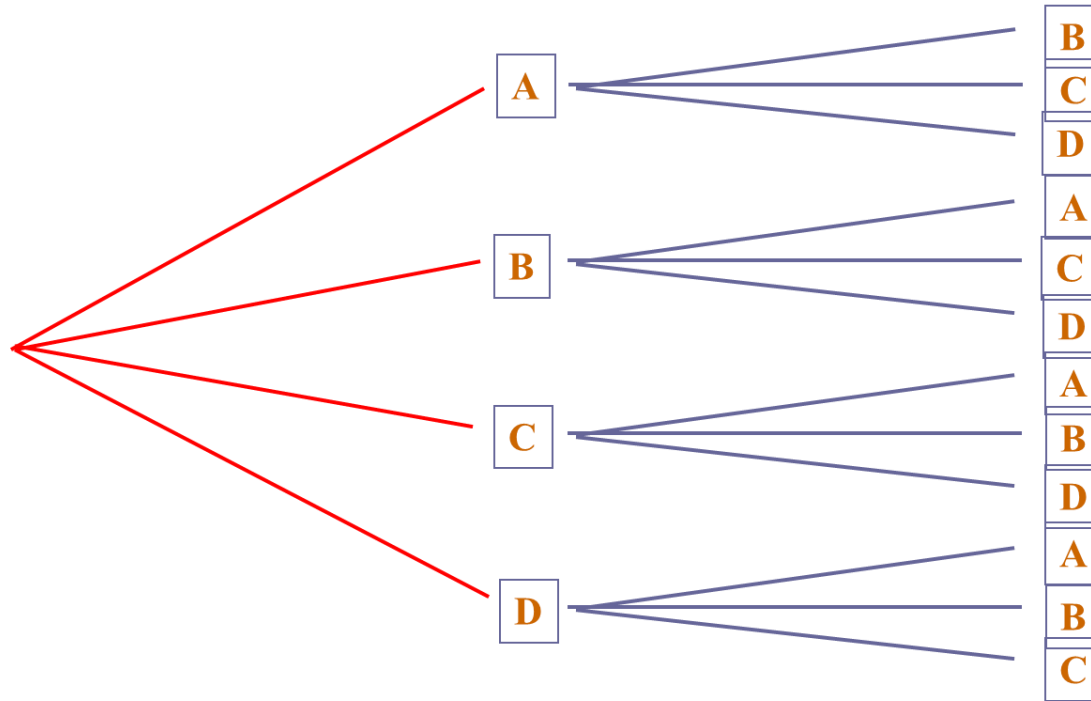
Sample Space: Roster Example

- Experiment: randomly select, without replacement, two families from the residents of Tiny Town
- Each ordered pair in the sample space is an elementary event, for example -- (D,C)

Family	Children in Household	Number of Automobiles
A	Yes	3
B	Yes	2
C	No	1
D	Yes	2

Listing of Sample Space
(A,B), (A,C), (A,D), (B,A), (B,C), (B,D), (C,A), (C,B), (C,D), (D,A), (D,B), (D,C)

Sample Space: Tree Diagram for Random Sample of Two Families



Sample Space: Set Notation for Random Sample of Two Families

- $S = \{(x,y) \mid x \text{ is the family selected on the first draw, and } y \text{ is the family selected on the second draw}\}$
- Concise description of large sample spaces

Sample Space

- Useful for discussion of general principles and concepts

Listing of Sample Space

(A,B), (A,C), (A,D),
(B,A), (B,C), (B,D),
(C,A), (C,B), (C,D),
(D,A), (D,B), (D,C)

Venn Diagram



Union of Sets

- The union of two sets contains an instance of each element of the two sets.

$$X = \{1,4,7,9\}$$

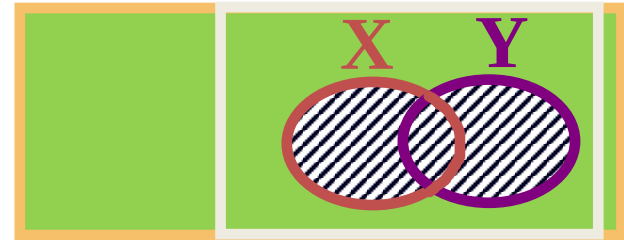
$$Y = \{2,3,4,5,6\}$$

$$X \cup Y = \{1,2,3,4,5,6,7,9\}$$

$$C = \{IBM, DEC, Apple\}$$

$$F = \{Apple, Grape, Lime\}$$

$$C \cup F = \{IBM, DEC, Apple, Grape, Lime\}$$



Intersection of Sets

- The intersection of two sets contains only those element common to the

$$X = \{1,4,7,9\}$$

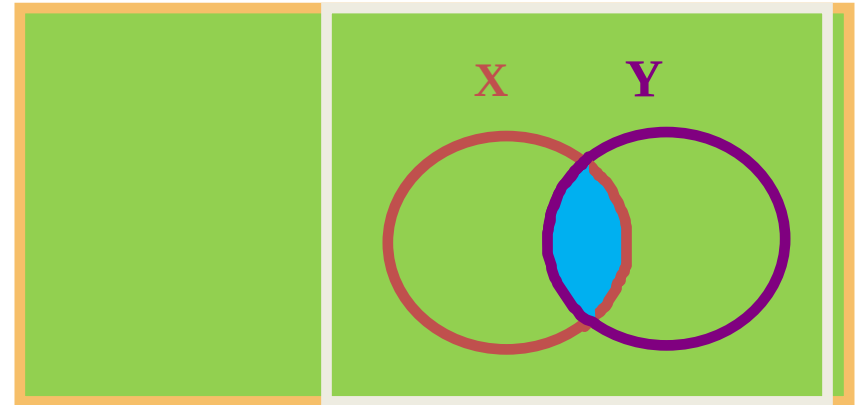
$$Y = \{2,3,4,5,6\}$$

$$X \cap Y = \{4\}$$

$$C = \{IBM, DEC, Apple\}$$

$$F = \{Apple, Grape, Lime\}$$

$$C \cap F = \{Apple\}$$



Mutually Exclusive Events

- Events with no common outcomes
- Occurrence of one event precludes the occurrence of the other event

$$C = \{IBM, DEC, Apple\}$$

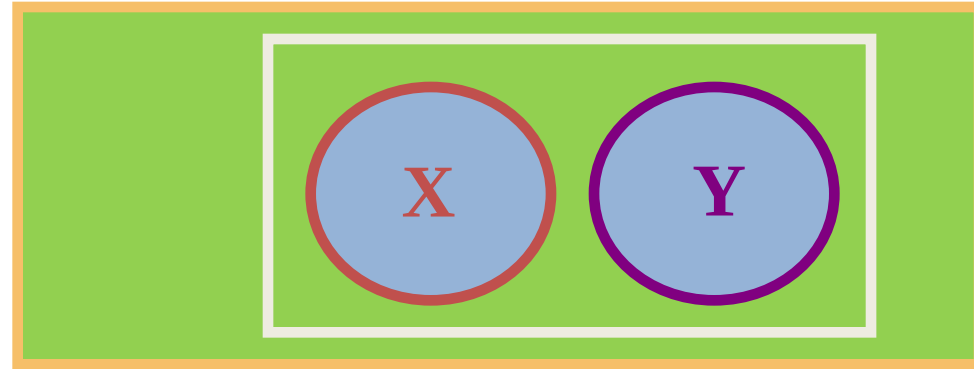
$$F = \{Grape, Lime\}$$

$$C \cap F = \{ \}$$

$$X = \{1, 7, 9\}$$

$$Y = \{2, 3, 4, 5, 6\}$$

$$X \cap Y = \{ \}$$



$$P(X \cap Y) = 0$$

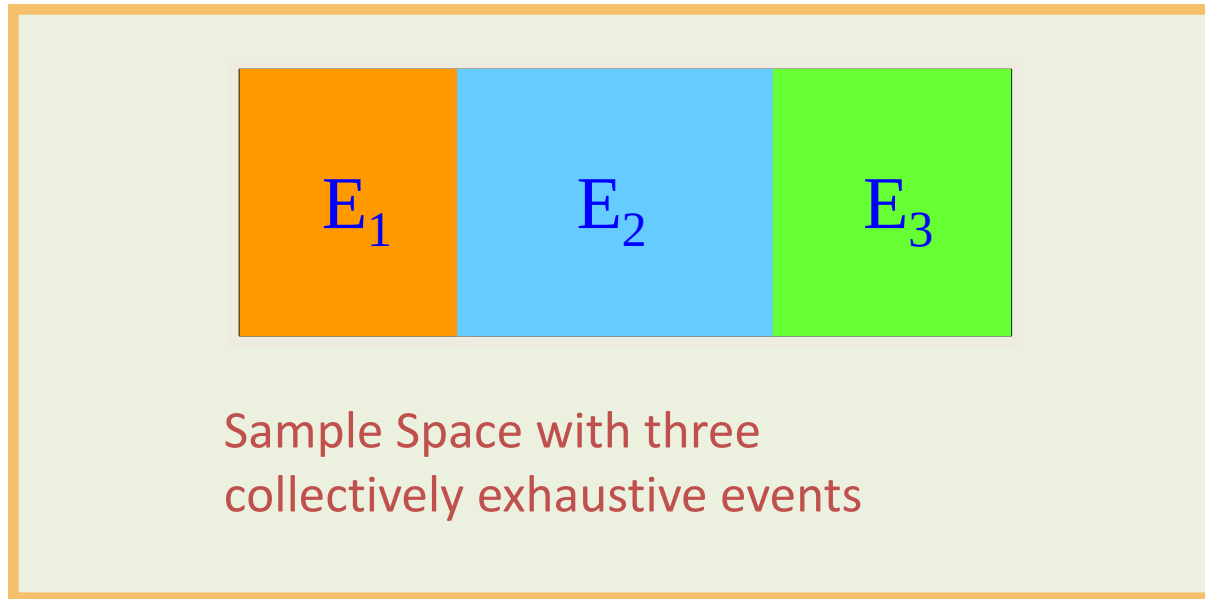
Independent Events

- Occurrence of one event does not affect the occurrence or nonoccurrence of the other event
- The conditional probability of X given Y is equal to the marginal probability of X.
- The conditional probability of Y given X is equal to the marginal probability of Y.

$$P(X|Y) = P(X) \text{ and } P(Y|X) = P(Y)$$

Collectively Exhaustive Events

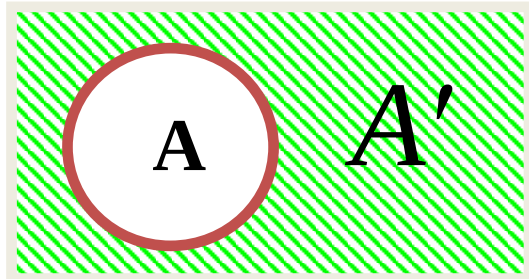
- Contains all elementary events for an experiment



Complementary Events

- All elementary events not in the event 'A' are in its complementary event.

Sample
Space



$$P(\text{Sample Space}) = 1$$

$$P(A') = 1 - P(A)$$

Counting the Possibilities

- mn Rule
- Sampling from a Population with Replacement
- Combinations: Sampling from a Population without Replacement

mn Rule

- If an operation can be done **m** ways and a second operation can be done **n** ways, then there are **mn** ways for the two operations to occur in order.
- This rule is easily extend to k stages, with a number of ways equal to $n_1 \cdot n_2 \cdot n_3 \dots n_k$
- Example: Toss two coins . The total umber of simple events is **2 x 2 =4**

Sampling from a Population with Replacement

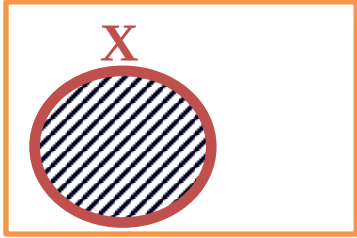
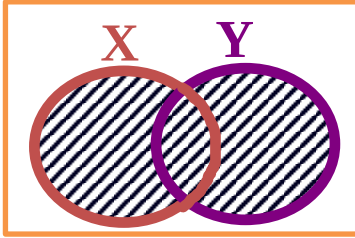
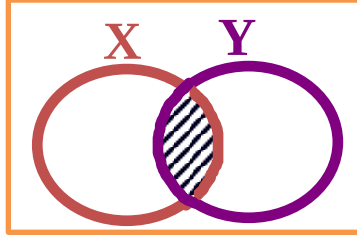
- A tray contains 1,000 individual tax returns. If 3 returns are randomly selected **with replacement** from the tray, how many possible samples are there?
- $(N)^n = (1,000)^3 = 1,000,000,000$

Combinations

- A tray contains 1,000 individual tax returns. If 3 returns are randomly selected **without replacement** from the tray, how many possible samples are there?

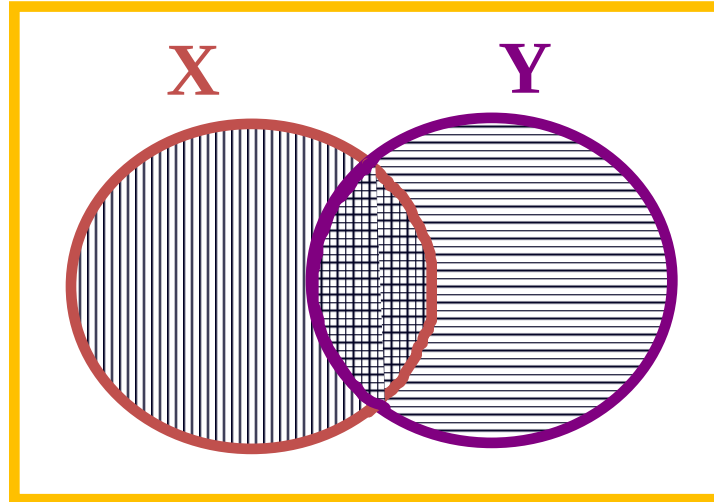
$$\binom{N}{n} = \frac{N!}{n!(N-n)!} = \frac{1000!}{3!(1000-3)!} = 166,167,000$$

Four Types of Probability

Marginal	Union	Joint	Conditional
$P(X)$ The probability of X occurring 	$P(X \cup Y)$ The probability of X or Y occurring 	$P(X \cap Y)$ The probability of X and Y occurring 	$P(X Y)$ The probability of X occurring given that Y has occurred 

General Law of Addition

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$



Design for improving productivity?



Problem

- A company conducted a survey for the American Society of Interior Designers in which workers were asked which changes in office design would increase productivity.
- Respondents were allowed to answer more than one type of design change.

Reducing noise would increase productivity	70 %
More storage space would increase productivity	67 %

Problem

- If one of the survey respondents was randomly selected and asked what office design changes would increase worker productivity,
 - what is the probability that this person would select reducing noise *or* more storage space?

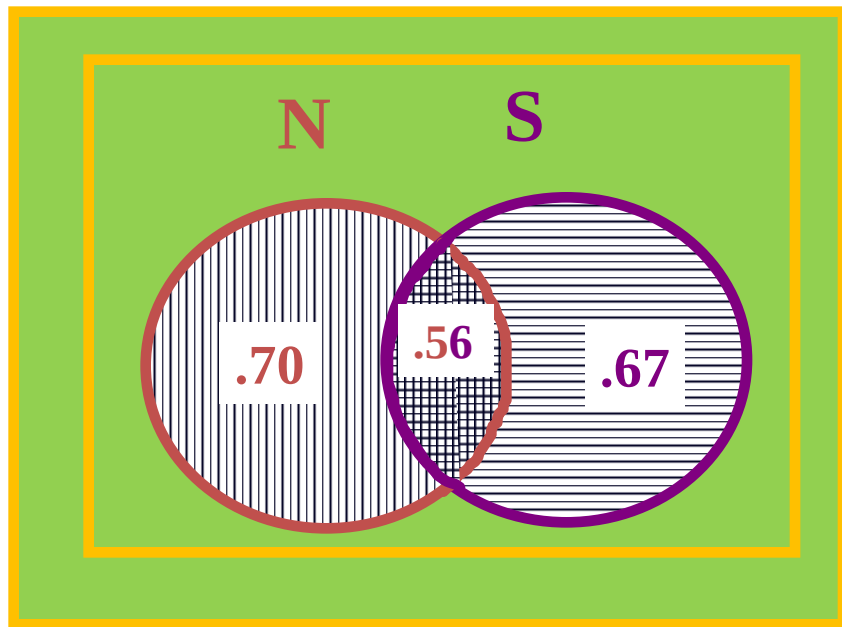
Solution

- Let N represent the event “reducing noise.”
- Let S represent the event “more storage/ filing space.”
- The probability of a person responding with N or S can be symbolized statistically as a union probability by using the law of addition.

$$P(N \cup S)$$

General Law of Addition -- Example

$$P(N \cup S) = P(N) + P(S) - P(N \cap S)$$



$$P(N) = .70$$

$$P(S) = .67$$

$$P(N \cap S) = .56$$

$$\begin{aligned} P(N \cup S) &= .70 + .67 - .56 \\ &= 0.81 \end{aligned}$$

Office Design Problem

Probability Matrix

		Increase Storage Space		
Noise Reduction		Yes	No	Total
	Yes	.56	.14	.70
	No	.11	.19	.30
	Total	.67	.33	1.00

Joint Probability Using a Contingency Table

Event	Event		Total
	B_1	B_2	
A_1	$P(A_1 \text{ and } B_1)$	$P(A_1 \text{ and } B_2)$	$P(A_1)$
A_2	$P(A_2 \text{ and } B_1)$	$P(A_2 \text{ and } B_2)$	$P(A_2)$
Total	$P(B_1)$	$P(B_2)$	1

Joint Probabilities

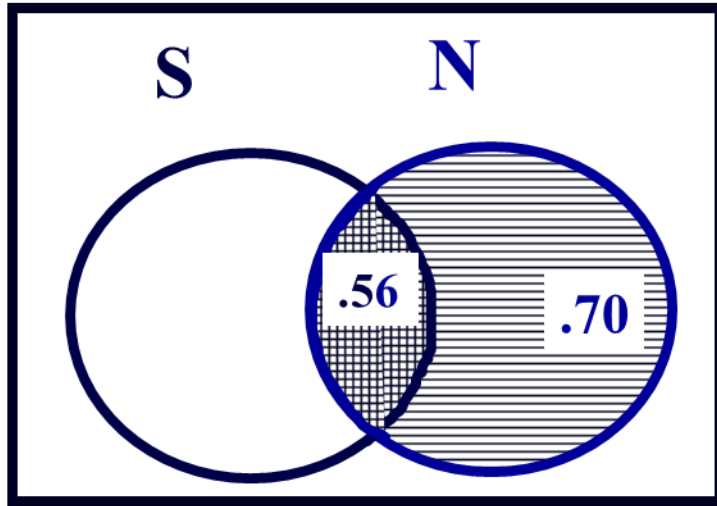
Marginal (Simple) Probabilities

Office Design Problem - Probability Matrix

		Increase Storage Space		Total
		Yes	No	
Noise Reduction	Yes	.56	.14	.70
	No	.11	.19	.30
	Total	.67	.33	1.00

$$\begin{aligned}P(N \cup S) &= P(N) + P(S) - P(N \cap S) \\&= .70 + .67 - .56 \\&= .81\end{aligned}$$

Law of Conditional Probability



$$P(N) = .70$$

$$P(N \cap S) = .56$$

$$\begin{aligned} P(S|N) &= \frac{P(N \cap S)}{P(N)} \\ &= \frac{.56}{.70} \\ &= .80 \end{aligned}$$

Office Design Problem

		Increase Storage Space		
		Yes	No	Total
Noise Reduction	Yes	.56	.14	.70
	No	.11	.19	.30
Total		.67	.33	1.00

$$P(\bar{N} | S) = \frac{P(\bar{N} \cap S)}{P(S)} = \frac{.11}{.67} \\ = .164$$

Problem

- A company data reveal that 155 employees worked one of four types of positions.
- Shown here again is the raw values matrix (also called a contingency table) with the frequency counts for each category and for subtotals and totals containing a breakdown of these employees by type of position and by sex.

Contingency Table

COMPANY HUMAN RESOURCE DATA

		<i>Sex</i>		
		<i>Male</i>	<i>Female</i>	
<i>Type of Position</i>	<i>Managerial</i>	8	3	11
	<i>Professional</i>	31	13	44
	<i>Technical</i>	52	17	69
	<i>Clerical</i>	9	22	31
		100	55	155

Solution

- If an employee of the company is selected randomly, what is the probability that the employee is female or a professional worker?

$$P(F \cup P) = P(F) + P(P) - P(F \cap P)$$

$$P(F \cup P) = .355 + .284 - .084 = .555.$$

Problem

- Shown here are the raw values matrix and corresponding probability matrix for the results of a national survey of 200 executives who were asked to identify the geographic locale of their company and their company's industry type.
- The executives were only allowed to select one locale and one industry type.



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Lecture 7: Introduction to Probability-II

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Problem

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Problem

- Shown here are the raw values matrix and corresponding probability matrix for the results of a national survey of 200 executives who were asked to identify the geographic locale of their company and their company's industry type.
- The executives were only allowed to select one locale and one industry type.

RAW VALUES MATRIX

Geographic Location

Industry Type	Geographic Location				
	Northeast	Southeast	Midwest	West	
	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	
<i>Finance A</i>	24	10	8	14	56
<i>Manufacturing B</i>	30	6	22	12	70
<i>Communications C</i>	28	18	12	16	74
	82	34	42	42	200

Questions

- a.** What is the probability that the respondent is from the Midwest (F)?
- b.** What is the probability that the respondent is from the communications industry (C) or from the Northeast (D)?
- c.** What is the probability that the respondent is from the Southeast (E) or from the finance industry (A)?

PROBABILITY MATRIX

		Geographic Location				
		Northeast	Southeast	Midwest	West	
		D	E	F	G	
Industry Type	Finance A	.12	.05	.04	.07	.28
	Manufacturing B	.15	.03	.11	.06	.35
	Communications C	.14	.09	.06	.08	.37
		.41	.17	.21	.21	1.00

Mutually Exclusive Events

Type of Position	Gender		Total
	Male	Female	
Managerial	8	3	11
Professional	31	13	44
Technical	52	17	69
Clerical	9	22	31
Total	100	55	155

$$\begin{aligned}P(T \cup C) &= P(T) + P(C) \\&= \frac{69}{155} + \frac{31}{155} \\&= .645\end{aligned}$$

Mutually Exclusive Events

Type of Position	Gender		Total
	Male	Female	
Managerial	8	3	11
Professional	31	13	44
Technical	52	17	69
Clerical	9	22	31
Total	100	55	155

$$P(P \cup C) = P(P) + P(C)$$

$$= \frac{44}{155} + \frac{31}{155}$$

$$=.484$$

Law of Multiplication

$$P(X \cap Y) = P(X) \cdot P(Y|X) = P(Y) \cdot P(X|Y)$$

Problem

- A company has 140 employees, of which 30 are supervisors.
- Eighty of the employees are married, and 20% of the married employees are supervisors.
- If a company employee is randomly selected, what is the probability that the employee is married and is a supervisor?

		Married		
		Y	N	Sub total
Supervisor	Y	0.1143		30
	N			110
	Sub total	80	60	140

$$P(M) = \frac{80}{140} = 0.5714$$

$$P(S|M) = 0.20$$

$$\begin{aligned} P(M \cap S) &= P(M) \cdot P(S|M) \\ &= (0.5714)(0.20) = 0.1143 \end{aligned}$$

Law of Multiplication

Probability Matrix of Employees

		Married		
		Yes	No	Total
Supervisor	Yes	.1143	.1000	.2143
	No	.4571	.3286	.7857
	Total	.5714	.4286	1.00

$$P(\bar{S}) = 1 - P(S)$$

$$= 1 - 0.2143 = 0.7857$$

$$P(\bar{M} \cap \bar{S}) = P(\bar{S}) - P(M \cap \bar{S})$$

$$= 0.7857 - 0.4571 = 0.3286$$

$$P(M \cap \bar{S}) = P(M) - P(M \cap S)$$

$$= 0.5714 - 0.1143 = 0.4571$$

$$P(\bar{M} \cap S) = P(S) - P(M \cap S)$$

$$= 0.2143 - 0.1143 = 0.1000$$

$$P(\bar{M}) = 1 - P(M)$$

$$= 1 - 0.5714 = 0.4286$$

Special Law of Multiplication for Independent Events

- General Law

$$P(X \cap Y) = P(X) \cdot P(Y | X) = P(Y) \cdot P(X | Y)$$

- Special Law

If events X and Y are independent,
 $P(X) = P(X | Y)$, and $P(Y) = P(Y | X)$.
Consequently,
 $P(X \cap Y) = P(X) \cdot P(Y)$

Law of Conditional Probability

- The conditional probability of X given Y is the joint probability of X and Y divided by the marginal probability of Y.

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{P(Y|X) \cdot P(X)}{P(Y)}$$

Conditional Probability

- A conditional probability is the probability of one event, given that another event has occurred:

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)}$$



The conditional probability of A given that B has occurred

$$P(B | A) = \frac{P(A \text{ and } B)}{P(A)}$$



The conditional probability of B given that A has occurred

Where $P(A \text{ and } B)$ = joint probability of A and B

$P(A)$ = marginal probability of A

$P(B)$ = marginal probability of B

Computing Conditional Probability

- Of the cars on a used car lot, 70% have air conditioning (AC) and 40% have a CD player (CD). 20% of the cars have both.
- What is the probability that a car has a CD player, given that it has AC ?
- We want to find $P(\text{CD} \mid \text{AC})$.



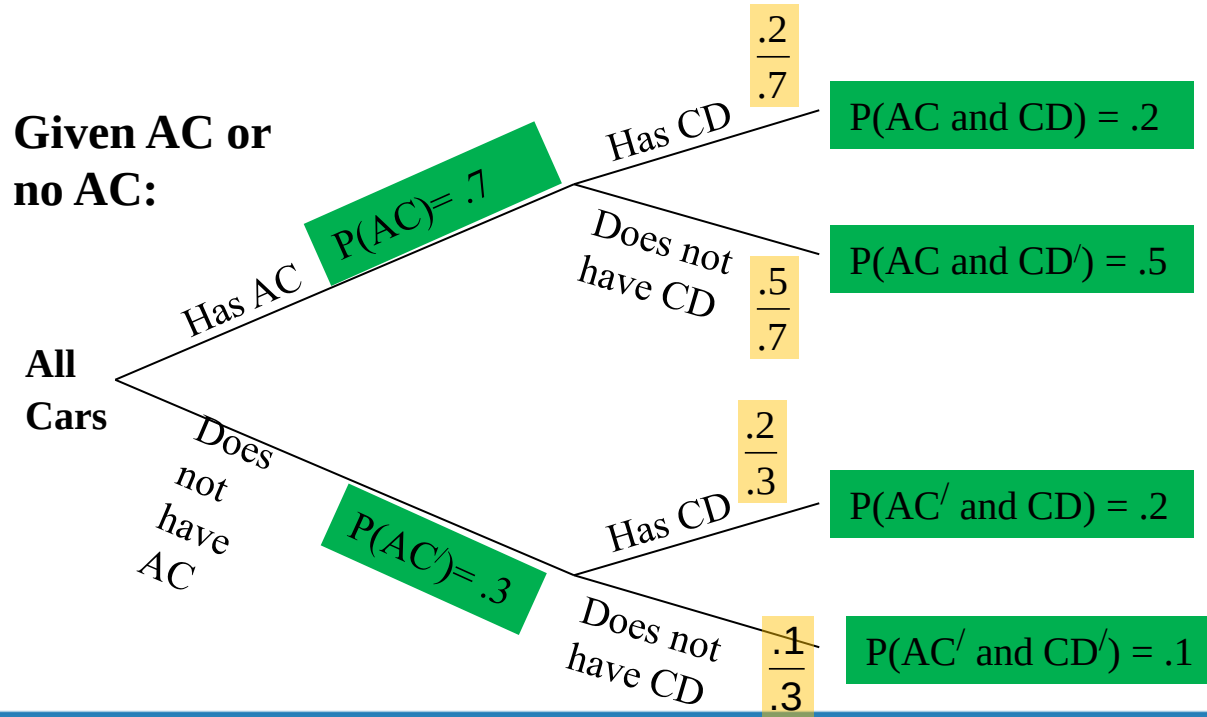
Computing Conditional Probability

	CD	No CD	Total
AC	0.2	0.5	0.7
No AC	0.2	0.1	0.3
Total	0.4	0.6	1.0

$$P(\text{CD} | \text{AC}) = \frac{P(\text{CD and AC})}{P(\text{AC})} = \frac{.2}{.7} = .2857$$

Given AC, we only consider the top row (70% of the cars). Of these, 20% have a CD player. 20% of 70% is about 28.57%.

Computing Conditional Probability: Decision Trees



Independent Events

- If X and Y are independent events, the occurrence of Y does not affect the probability of X occurring.
- If X and Y are independent events, the occurrence of X does not affect the probability of Y occurring.

If X and Y are independent events,

$$P(X|Y) = P(X), \text{ and}$$

$$P(Y|X) = P(Y).$$

Statistical Independence

- Two events are **independent** if and only if:

$$P(A | B) = P(A)$$

- Events A and B are independent when the probability of one event is not affected by the other event

Independent Events Demonstration

		Geographic Location				
		Northeast D	Southeast E	Midwest F	West G	
Finance	A	.12	.05	.04	.07	.28
Manufacturing	B	.15	.03	.11	.06	.35
Communications	C	.14	.09	.06	.08	.37
		.41	.17	.21	.21	1.00

Test the matrix for the 200 executive responses to determine whether industry type is independent of geographic location.

Independent Events Demonstration Contd...

$$P(A|G) = \frac{P(A \cap G)}{P(G)} = \frac{0.07}{0.21} = 0.33 \quad P(A) = 0.28$$

$$P(A|G) = 0.33 \neq P(A) = 0.28$$



Independent Events

	D	E	
A	8	12	20
B	20	30	50
C	6	9	15
	34	51	85

$$P(A|D) = \frac{8}{34} = .2353$$

$$P(A) = \frac{20}{85} = .2353$$

$$P(A|D) = P(A) = 0.2353$$

Revision of Probabilities: Bayes' Rule

- An extension to the conditional law of probabilities
- Enables revision of original probabilities with new information

$$P(X_i|Y) = \frac{P(Y|X_i)P(X_i)}{P(Y|X_1)P(X_1) + P(Y|X_2)P(X_2) + \cdots P(Y|X_n)P(X_n)}$$









Problem

- A particular type of printer ribbon is produced by only two companies, **Alamo Ribbon Company** and **South Jersey Products**.
- Suppose **Alamo produces 65%** of the ribbons and that **South Jersey produces 35%**.
- **Eight percent** of the ribbons produced by Alamo are defective and 12% of the South Jersey ribbons are defective
- A customer purchases a new ribbon. What is the probability that Alamo produced the ribbon? What is the probability that South Jersey produced the ribbon?

Revision of Probabilities with Bayes' Rule: Ribbon Problem

$$P(\text{Alamo}) = 0.65$$

$$P(\text{South Jersey}) = 0.35$$

$$P(d|\text{Alamo}) = 0.08$$

$$P(d|\text{South Jersey}) = 0.12$$

$$\begin{aligned} P(\text{Alamo}|d) &= \frac{P(d|\text{Alamo}) \cdot P(\text{Alamo})}{P(d|\text{Alamo}) \cdot P(\text{Alamo}) + P(d|\text{South Jersey}) \cdot P(\text{South Jersey})} \\ &= \frac{(0.08)(0.65)}{(0.08)(0.65) + (0.12)(0.35)} = 0.553 \end{aligned}$$

$$\begin{aligned} P(\text{South Jersey}|d) &= \frac{P(d|\text{South Jersey}) \cdot P(\text{South Jersey})}{P(d|\text{Alamo}) \cdot P(\text{Alamo}) + P(d|\text{South Jersey}) \cdot P(\text{South Jersey})} \\ &= \frac{(0.12)(0.35)}{(0.08)(0.65) + (0.12)(0.35)} = 0.447 \end{aligned}$$

Revision of Probabilities with Bayes' Rule: Ribbon Problem

Event	Prior Probability $P(E_i)$	Conditional Probability $P(d E_i)$	Joint Probability $P(E_i \cap d)$	Revised Probability $P(E_i d)$
Alamo	0.65	0.08	0.052	$\frac{0.052}{0.094}$ =0.553
South Jersey	0.35	0.12	$\frac{0.042}{0.094}$	$\frac{0.042}{0.094}$ =0.447

Revision of Probabilities with Bayes' Rule: Ribbon Problem



THANK YOU





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Data Analytics with Python

Lecture 8: Probability Distributions

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT
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Lecture Objectives

- Empirical Distribution
- Discrete Distributions
- Continuous Distributions

What is a distribution?

- Describes the 'shape' of a batch of numbers
- The characteristics of a distribution can sometimes be defined using a small number of numeric descriptors called 'parameters'

Why distribution?

- Can serve as a basis for standardized comparison of empirical distributions
- Can help us estimate confidence intervals for inferential statistics
- Form a basis for more advanced statistical methods
 - ‘fit’ between observed distributions and certain theoretical distributions is an assumption of many statistical procedures

Random variable

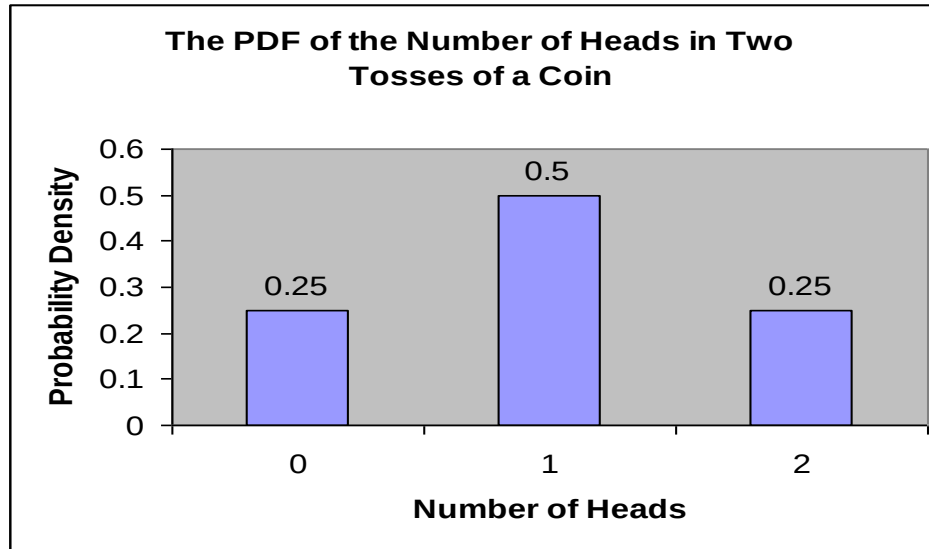
- A variable which contains the outcomes of a chance experiment
- “Quantifying the outcomes”
- Example $X = (1 = \text{Head}, 0 = \text{Tails})$
- A variable that can take on different values in the population according to some “random” mechanism
- Discrete
 - Distinct values, countable
 - Year
- Continuous
 - Mass

Probability Distributions

- The probability distribution function or probability density function (PDF) of a random variable X means the values taken by that random variable and their associated probabilities.
- PDF of a discrete r.v. (also known as PMF):
Example 1: Let the r.v. X be the number of heads obtained in two tosses of a coin.
Sample Space: $\{HH, HT, TH, TT\}$

PDF of Discrete r.v.

Number of Heads (X):	0	1	2	sum	
PDF ($P(X)$):		$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	1



Probability Distribution for the Random Variable X

A probability distribution for a discrete random variable X :

x	-8	-3	-1	0	1	4	6
$P(X = x)$	0.13	0.15	0.17	0.20	0.15	0.11	0.09

Find

a. $P(X \leq 0)$ 0.65

b. $P(-3 \leq X \leq 1)$ 0.67

Discrete Distribution -- Example

Distribution of Daily Crises	
Number of Crises	Probability
0	0.37
1	0.31
2	0.18
3	0.09
4	0.04
5	0.01



Requirements for a Discrete Probability Function

- Probabilities are between 0 and 1, inclusively
- Total of all probabilities equals 1

$$0 \leq P(X) \leq 1 \quad \text{for all } X$$

$$\sum_{\text{over all } x} P(X) = 1$$

Cumulative Distribution Function

- The CDF of a random variable X (defined as $F(X)$) is a graph associating all possible values, or the range of possible values with $P(X \leq x)$.
- CDFs always lie between 0 and 1 i.e., $0 \leq F(X_i) \leq 1$, Where $F(X_i)$ is the CDF.

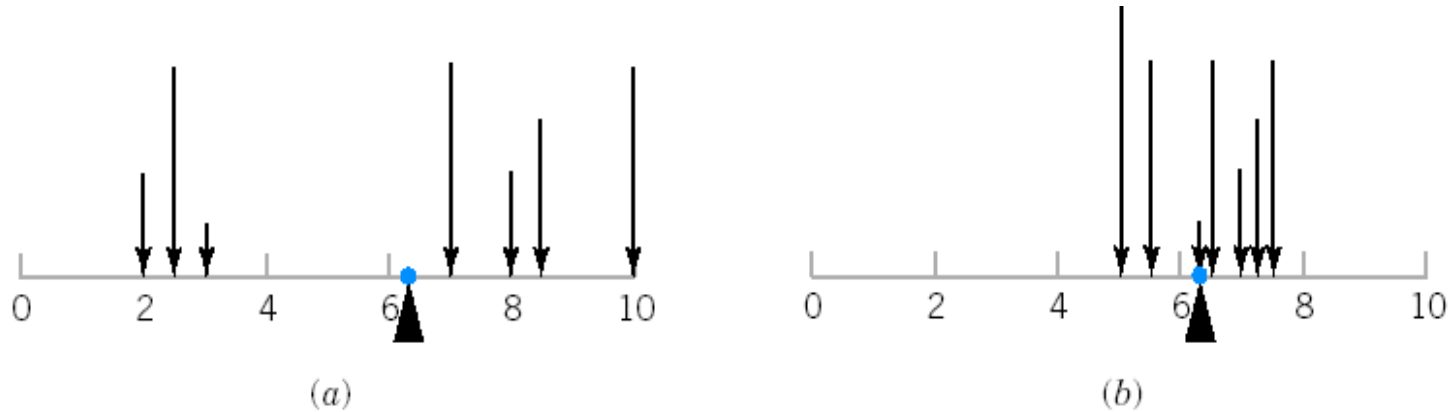
The Expected Value of X

Let X be a discrete rv with set of possible values D and pmf $p(x)$. The *expected value* or *mean value* of X , denoted

$E(X)$ or μ_X , is

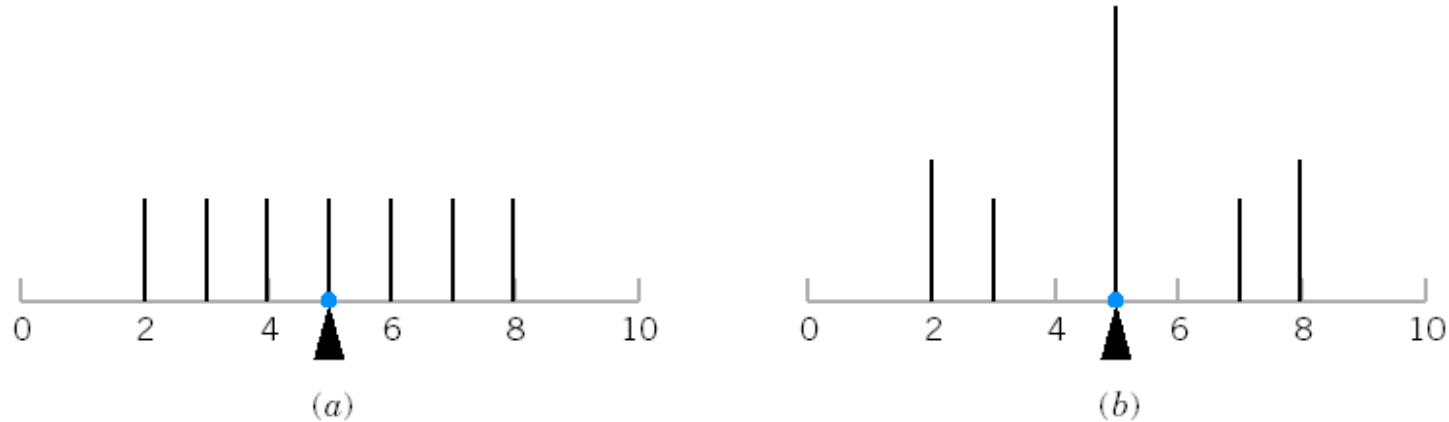
$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

Mean and Variance of a Discrete Random Variable



A probability distribution can be viewed as a loading with the mean equal to the balance point. Parts (a) and (b) illustrate equal means, but Part (a) illustrates a larger variance.

Mean and Variance of a Discrete Random Variable



The probability distribution illustrated in Parts (a) and (b) differ even though they have equal means and equal variances.

Example – Expected Value

- Use the data below to find out the expected number of credit cards that a customer to a retail outlet will possess.

x = # credit cards

x	$P(x=X)$
0	0.08
1	0.28
2	0.38
3	0.16
4	0.06
5	0.03
6	0.01

$$\begin{aligned} E(X) &= x_1p_1 + x_2p_2 + \dots + x_np_n \\ &= 0(.08) + 1(.28) + 2(.38) + 3(.16) \\ &\quad + 4(.06) + 5(.03) + 6(.01) \\ &= 1.97 \end{aligned}$$

About 2 credit cards

The Variance and Standard Deviation

Let X have pmf $p(x)$, and expected value μ . Then the variance of X , denoted $V(X)$

(or σ_X^2 or σ^2), is

$$V(X) = \sum_D (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2]$$

The standard deviation (SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

The quiz scores for a particular student are given below:

22, 25, 20, 18, 12, 20, 24, 20, 20, 25, 24, 25, 18

Find the variance and standard deviation.

Value	12	18	20	22	24	25
Frequency	1	2	4	1	2	3
Probability	.08	.15	.31	.08	.15	.23

$$\mu = 21$$

$$V(X) = p_1(x_1 - \mu)^2 + p_2(x_2 - \mu)^2 + \dots + p_n(x_n - \mu)^2$$

$$\sigma = \sqrt{V(X)}$$

$$V(X) = .08(12 - 21)^2 + .15(18 - 21)^2 + .31(20 - 21)^2 \\ + .08(22 - 21)^2 + .15(24 - 21)^2 + .23(25 - 21)^2$$

$$V(X) = 13.25$$

$$\sigma = \sqrt{V(X)} = \sqrt{13.25} \approx 3.64$$

Shortcut Formula for Variance

$$\begin{aligned} V(X) = \sigma^2 &= \left[\sum_D x^2 \cdot p(x) \right] - \mu^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

Mean of a Discrete Distribution

$$\mu = E(X) = \sum X \cdot P(X)$$

X	P(X)	X.P(X)
-1	.1	-.1
0	.2	.0
1	.4	.4
2	.2	.4
3	.1	<u>.3</u>
		1.0

Variance and Standard Deviation of a Discrete Distribution

$$\sigma^2 = \sum (X - \mu)^2 \cdot P(X) = 1.2$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{1.2} \cong 1.10$$

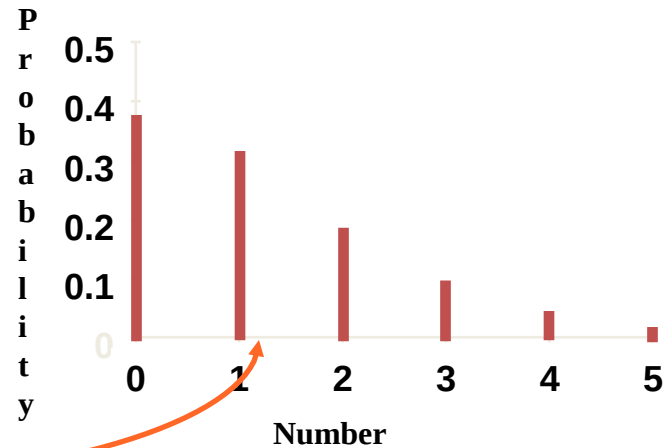
X	P(X)	$X - \mu$	$(X - \mu)^2$	$(X - \mu)^2 \cdot P(X)$
-1	.1	-2	4	.4
0	.2	-1	1	.2
1	.4	0	0	.0
2	.2	1	1	.2
3	.1	2	4	.4
				<u>1.2</u>

Mean of the Data Example

$$\mu = E(X) = \sum X \cdot P(X) = 1.15$$

X	P(X)	X•P(X)
0	.37	.00
1	.31	.31
2	.18	.36
3	.09	.27
4	.04	.16
5	.01	.05

1.15



Properties of Expected Value

1. $E(b) = b$, b is a constant.

2. $E(X + Y) = E(X) + E(Y)$.

3. $E\left(\frac{X}{Y}\right) \neq \frac{E(X)}{E(Y)}$.

4. $E(XY) \neq E(X)E(Y)$ unless they are independent.

5. $E(aX) = aE(X)$, a constant.

6. $E(aX + b) = aE(X) + b$, a and b are constants.

Properties of Variance

1. $\text{Var}(\text{constant}) = 0$
2. If X and Y are two independent random variables, then
 $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ and
 $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$
3. If b is a constant then $\text{Var}(b+X) = \text{Var}(X)$
4. If a is a constant then $\text{Var}(aX) = a^2\text{Var}(X)$
5. If a and b are constants then $\text{Var}(aX+b) = a^2\text{Var}(X)$
6. If X and Y are two independent random variables and a and b are constants then $\text{Var}(aX+bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$

Covariance

Covariance: For two discrete random variables X and Y with $E(X) = \mu_x$ and $E(Y) = \mu_y$, the covariance between X and Y is defined as

$$\text{Cov}(XY) = \sigma_{xy} = E(X - \mu_x) E(Y - \mu_y) = E(XY) - \mu_x \mu_y.$$

Covariance

- In general, the covariance between two random variables can be positive or negative.
- If two random variables move in the same direction, then the covariance will be positive, if they move in the opposite direction the covariance will be negative.

Properties:

- 1.If X and Y are independent random variables, their covariance is zero. Since $E(XY) = E(X)E(Y)$
2. $\text{Cov}(XX) = \text{Var}(X)$
3. $\text{Cov}(YY) = \text{Var}(Y)$

Correlation Coefficient

- The covariance tells the sign but not the magnitude about how strongly the variables are positively or negatively related. The correlation coefficient provides such measure of how strongly the variables are related to each other.
- For two random variables X and Y with $E(X) = \mu_x$ and $E(Y) = \mu_y$, the correlation coefficient is defined as

$$\rho_{xy} = \frac{\text{Cov}(XY)}{\sigma_x \sigma_y} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$



Thank You





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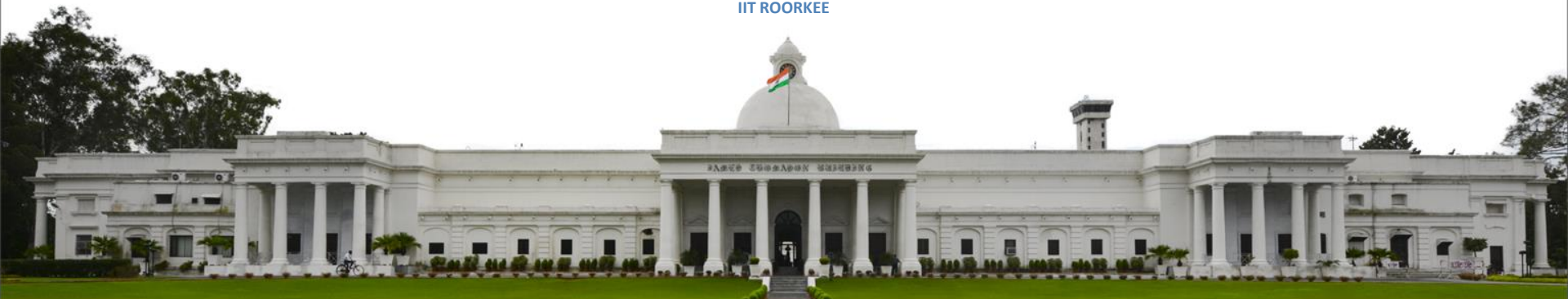


NPTEL ONLINE
CERTIFICATION COURSE

Data Analytics with Python

Lecture 9: Probability Distributions-II

Dr. A. Ramesh
DEPARTMENT OF MANAGEMENT
IIT ROORKEE



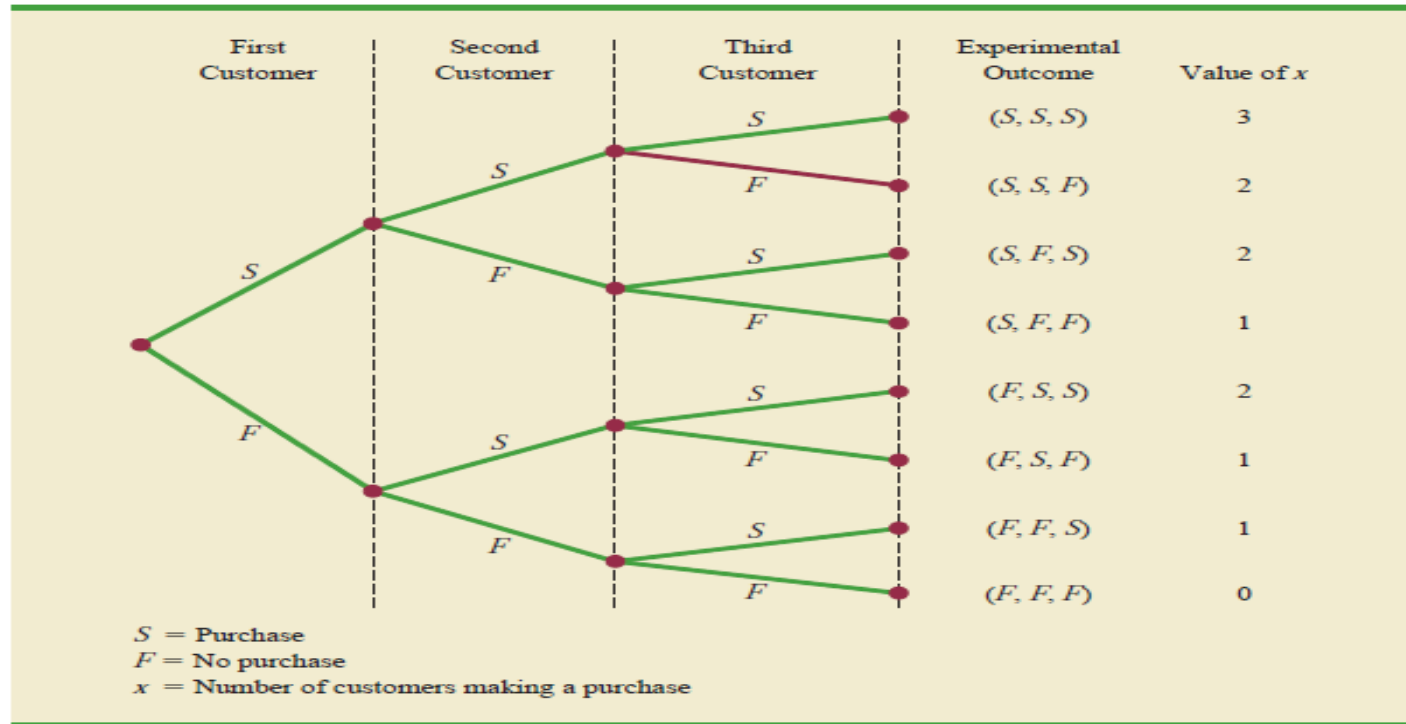
Some Special Distributions

- Discrete
 - Binomial
 - Poisson
 - Hyper geometric
- Continuous
 - Uniform
 - Exponential
 - Normal

Binomial Distribution

- Let us consider the purchase decisions of the next three customers who enter a store.
- On the basis of past experience, the store manager estimates the probability that any one customer will make a purchase is **.30**.
- What is the probability that **two** of the next **three** customers will make a purchase?

Tree diagram for the Martin clothing store problem



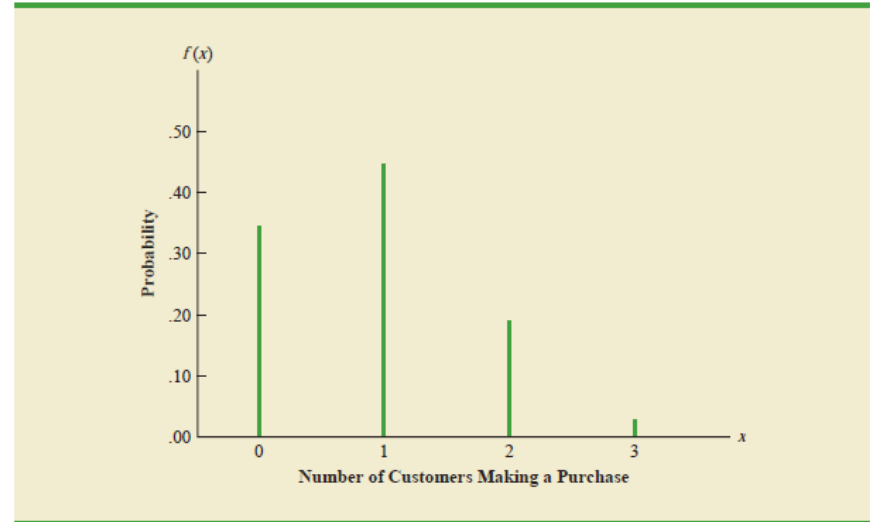
Trial Outcomes

Trial Outcomes

1st Customer	2nd Customer	3rd Customer	Experimental Outcome	Probability of Experimental Outcome
Purchase	Purchase	No purchase	(S, S, F)	$pp(1 - p) = p^2(1 - p)$ $= (.30)^2(.70) = .063$
Purchase	No purchase	Purchase	(S, F, S)	$p(1 - p)p = p^2(1 - p)$ $= (.30)^2(.70) = .063$
No purchase	Purchase	Purchase	(F, S, S)	$(1 - p)pp = p^2(1 - p)$ $= (.30)^2(.70) = .063$

Graphical representation of the probability distribution for the number of customers making a purchase

x	P(x)
0	$0.7 \times 0.7 \times 0.7 = 0.343$
1	$0.3 \times 0.7 \times 0.7 + 0.7 \times 0.3 \times 0.7 + 0.7 \times 0.7 \times 0.3 = 0.441$
2	0.189
3	0.027



Binomial Distribution- Assumptions

- Experiment involves n identical trials
 - Each trial has exactly two possible outcomes: success and failure
 - Each trial is independent of the previous trials
 - p is the probability of a success on any one trial
- $q = (1-p)$ is the probability of a failure on any one trial
- p and q are constant throughout the experiment
 - X is the number of successes in the n trials

Binomial Distribution

- Probability function

$$P(X) = \frac{n!}{X!(n-X)!} p^X \cdot q^{n-X} \quad \text{for } 0 \leq X \leq n$$

- Mean value

$$\mu = n \cdot p$$

- Variance and standard deviation

$$\begin{aligned}\sigma^2 &= n \cdot p \cdot q \\ \sigma &= \sqrt{\sigma^2} = \sqrt{n \cdot p \cdot q}\end{aligned}$$

Binomial Table

SELECTED VALUES FROM THE BINOMIAL PROBABILITY TABLE

EXAMPLE: $n = 10, x = 3, p = .40; f(3) = .2150$

n	x	p									
		.05	.10	.15	.20	.25	.30	.35	.40	.45	.50
9	0	.6302	.3874	.2316	.1342	.0751	.0404	.0207	.0101	.0046	.0020
	1	.2985	.3874	.3679	.3020	.2253	.1556	.1004	.0605	.0339	.0176
	2	.0629	.1722	.2597	.3020	.3003	.2668	.2162	.1612	.1110	.0703
	3	.0077	.0446	.1069	.1762	.2336	.2668	.2716	.2508	.2119	.1641
	4	.0006	.0074	.0283	.0661	.1168	.1715	.2194	.2508	.2600	.2461
	5	.0000	.0008	.0050	.0165	.0389	.0735	.1181	.1672	.2128	.2461
	6	.0000	.0001	.0006	.0028	.0087	.0210	.0424	.0743	.1160	.1641
	7	.0000	.0000	.0000	.0003	.0012	.0039	.0098	.0212	.0407	.0703
	8	.0000	.0000	.0000	.0000	.0001	.0004	.0013	.0035	.0083	.0176
	9	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0003	.0008	.0020
10	0	.5987	.3487	.1969	.1074	.0563	.0282	.0135	.0060	.0025	.0010
	1	.3151	.3874	.3474	.2684	.1877	.1211	.0725	.0403	.0207	.0098
	2	.0746	.1937	.2759	.3020	.2816	.2335	.1757	.1209	.0763	.0439
	3	.0105	.0574	.1298	.2013	.2503	.2668	.2522	.2150	.1665	.1172
	4	.0010	.0112	.0401	.0881	.1460	.2001	.2377	.2508	.2384	.2051
	5	.0001	.0015	.0085	.0264	.0584	.1029	.1536	.2007	.2340	.2461
	6	.0000	.0001	.0012	.0055	.0162	.0368	.0689	.1115	.1596	.2051
	7	.0000	.0000	.0001	.0008	.0031	.0090	.0212	.0425	.0746	.1172
	8	.0000	.0000	.0000	.0001	.0004	.0014	.0043	.0106	.0229	.0439
	9	.0000	.0000	.0000	.0000	.0000	.0001	.0005	.0016	.0042	.0098
	10	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0003	.0010

Mean and Variance

- Suppose that for the next month the Clothing Store forecasts 1000 customers will enter the store.
- What is the expected number of customers who will make a purchase?
- The answer is $\mu = np = (1000)(.3) = 300$.
- For the next 1000 customers entering the store, the variance and standard deviation for the number of customers who will make a purchase are

$$\sigma^2 = np(1 - p) = 1000(.3)(.7) = 210$$
$$\sigma = \sqrt{210} = 14.49$$

Poisson Distribution

- Describes discrete occurrences over a continuum or interval
- A discrete distribution
- Describes rare events
- Each occurrence is independent any other occurrences.
- The number of occurrences in each interval can vary from zero to infinity.
- The expected number of occurrences must hold constant throughout the experiment.

Poisson Distribution: Applications

- **Arrivals at queuing systems**
 - airports -- people, airplanes, automobiles, baggage
 - banks -- people, automobiles, loan applications
 - computer file servers -- read and write operations
- **Defects in manufactured goods**
 - number of defects per 1,000 feet of extruded copper wire
 - number of blemishes per square foot of painted surface
 - number of errors per typed page

Poisson Distribution

- Probability function

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!} \quad \text{for } X = 0, 1, 2, 3, \dots$$

where:

λ = long – run average

$e = 2.718282\dots$ (the base of natural logarithms)

Mean value

$$\lambda$$

Variance

$$\lambda$$

Standard deviation

$$\sqrt{\lambda}$$

Poisson Distribution: Example

$\lambda = 3.2$ customers/4 minutes

$X = 10$ customers/8 minutes

Adjusted λ

$\lambda = 6.4$ customers/8 minutes

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!}$$

$$P(X=10) = \frac{6.4^{10} e^{-6.4}}{10!} = 0.0528$$

$\lambda = 3.2$ customers/4 minutes

$X = 6$ customers/8 minutes

Adjusted λ

$\lambda = 6.4$ customers/8 minutes

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!}$$

$$P(X=6) = \frac{6.4^6 e^{-6.4}}{6!} = 0.1586$$

Poisson Probability Table

Example: $\mu = 10, x = 5; f(5) = .0378$

	μ									
x	9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	10
0	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0000
1	.0010	.0009	.0009	.0008	.0007	.0007	.0006	.0005	.0005	.0005
2	.0046	.0043	.0040	.0037	.0034	.0031	.0029	.0027	.0025	.0023
3	.0140	.0131	.0123	.0115	.0107	.0100	.0093	.0087	.0081	.0076
4	.0319	.0302	.0285	.0269	.0254	.0240	.0226	.0213	.0201	.0189
5	.0581	.0555	.0530	.0506	.0483	.0460	.0439	.0418	.0398	.0378
6	.0881	.0851	.0822	.0793	.0764	.0736	.0709	.0682	.0656	.0631
7	.1145	.1118	.1091	.1064	.1037	.1010	.0982	.0955	.0928	.0901
8	.1302	.1286	.1269	.1251	.1232	.1212	.1191	.1170	.1148	.1126
9	.1317	.1315	.1311	.1306	.1300	.1293	.1284	.1274	.1263	.1251
10	.1198	.1210	.1219	.1228	.1235	.1241	.1245	.1249	.1250	.1251
11	.0991	.1012	.1031	.1049	.1067	.1083	.1098	.1112	.1125	.1137
12	.0752	.0776	.0799	.0822	.0844	.0866	.0888	.0908	.0928	.0948
13	.0526	.0549	.0572	.0594	.0617	.0640	.0662	.0685	.0707	.0729
14	.0342	.0361	.0380	.0399	.0419	.0439	.0459	.0479	.0500	.0521
15	.0208	.0221	.0235	.0250	.0265	.0281	.0297	.0313	.0330	.0347
16	.0118	.0127	.0137	.0147	.0157	.0168	.0180	.0192	.0204	.0217
17	.0063	.0069	.0075	.0081	.0088	.0095	.0103	.0111	.0119	.0128
18	.0032	.0035	.0039	.0042	.0046	.0051	.0055	.0060	.0065	.0071
19	.0015	.0017	.0019	.0021	.0023	.0026	.0028	.0031	.0034	.0037
20	.0007	.0008	.0009	.0010	.0011	.0012	.0014	.0015	.0017	.0019
21	.0003	.0003	.0004	.0004	.0005	.0006	.0006	.0007	.0008	.0009
22	.0001	.0001	.0002	.0002	.0002	.0002	.0003	.0003	.0004	.0004
23	.0000	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0002	.0002
24	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0001	.0001

The Hypergeometric Distribution

- The **binomial distribution** is applicable when selecting from a finite population with replacement or from an infinite population without replacement.
- The hypergeometric distribution is applicable when selecting from a **finite population without replacement**.



Hyper Geometric Distribution

- Sampling without replacement from a finite population
- The number of objects in the population is denoted N .
- Each trial has exactly two possible outcomes, success and failure.
- Trials are not independent
- X is the number of successes in the n trials
- The binomial is an acceptable approximation, if $N/10 > n$ Otherwise it is not.



Hypergeometric Distribution

- Probability function
 - N is population size
 - n is sample size
 - A is number of successes in population
 - x is number of successes in sample
- Mean Value
- Variance and standard deviation

$$P(x) = \frac{{}_A C_x {}_{N-A} C_{n-x}}{{}_N C_n}$$

$$\mu = \frac{A \cdot n}{N}$$

$$\sigma^2 = \frac{A(N-A)n(N-n)}{N^2(N-1)}$$
$$\sigma = \sqrt{\sigma^2}$$

The Hypergeometric Distribution Example

- Different computers are checked from 10 in the department. 4 of the 10 computers have illegal software loaded.
- What is the probability that 2 of the 3 selected computers have illegal software loaded?
- So, $N = 10$, $n = 3$, $A = 4$, $X = 2$

$$P(X = 2) = \frac{\binom{A}{X} \binom{N-A}{n-X}}{\binom{N}{n}} = \frac{\binom{4}{2} \binom{6}{1}}{\binom{10}{3}} = \frac{(6)(6)}{120} = 0.3$$

- The probability that 2 of the 3 selected computers have illegal software loaded is .30, or 30%.

Continuous Probability Distributions

- A continuous random variable is a variable that can assume any value on a continuum (can assume an uncountable number of values)
 - thickness of an item
 - time required to complete a task
 - temperature of a solution
 - height
- These can potentially take on any value, depending only on the ability to measure precisely and accurately.



Continuous Distributions

- Uniform
- Normal
- Exponential



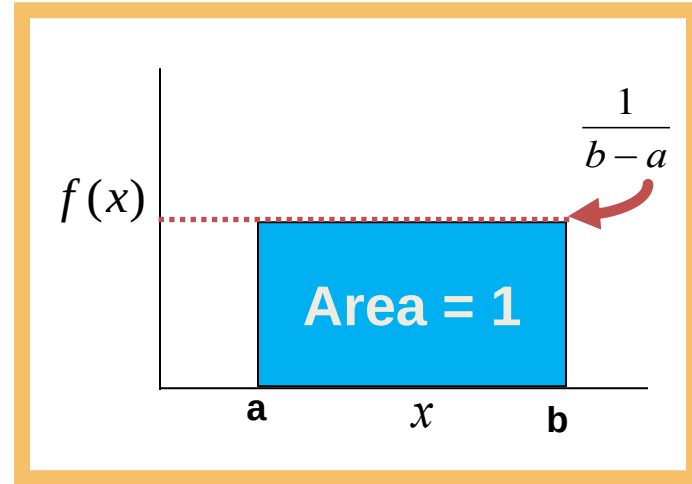
The Uniform Distribution

- The uniform distribution is a probability distribution that has equal probabilities for all possible outcomes of the random variable
- Because of its shape it is also called a rectangular distribution



Uniform Distribution

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for all other values} \end{cases}$$



Uniform Distribution: Mean and Standard Deviation

Mean

$$\mu = \frac{a + b}{2}$$

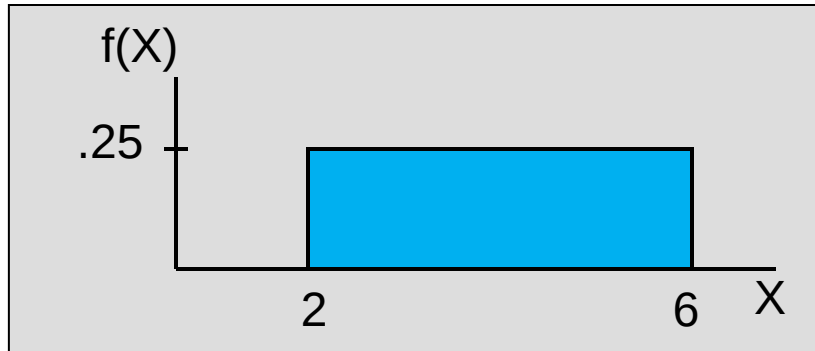
Standard Deviation

$$\sigma = \frac{b - a}{\sqrt{12}}$$

The Uniform Distribution

Example: Uniform probability distribution over the range $2 \leq X \leq 6$:

$$f(X) = \frac{1}{6 - 2} = .25 \quad \text{for } 2 \leq X \leq 6$$

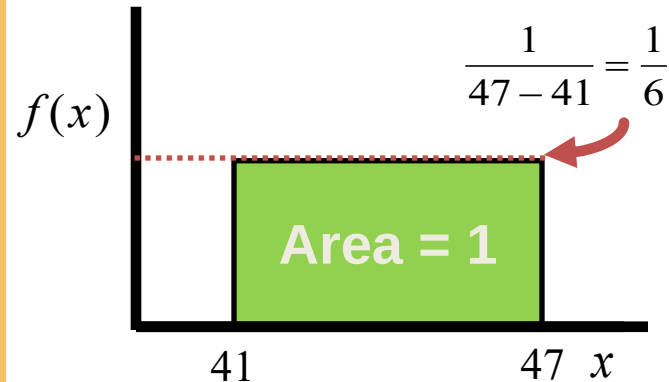


$$\mu = \frac{a + b}{2} = \frac{2 + 6}{2} = 4$$

$$\sigma = \sqrt{\frac{(b - a)^2}{12}} = \sqrt{\frac{(6 - 2)^2}{12}} = 1.1547$$

Uniform Distribution Example

$$f(x) = \begin{cases} \frac{1}{47-41} & \text{for } 41 \leq x \leq 47 \\ 0 & \text{for all other values} \end{cases}$$



Uniform Distribution: Mean and Standard Deviation

Mean

$$\mu = \frac{a + b}{2}$$

Mean

$$\mu = \frac{41+47}{2} = \frac{88}{2} = 44$$

Standard Deviation

$$\sigma = \frac{b - a}{\sqrt{12}}$$

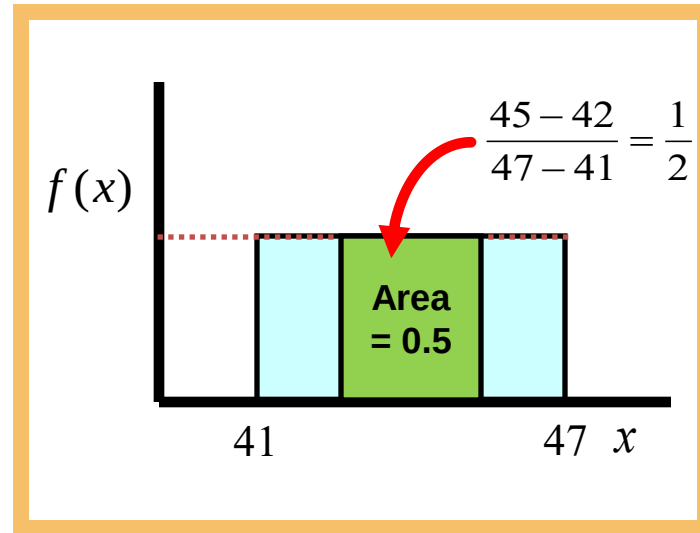
Standard Deviation

$$\sigma = \frac{47 - 41}{\sqrt{12}} = \frac{6}{3.464} = 1.732$$

Uniform Distribution Probability

$$P(x_1 \leq X \leq x_2) = \frac{x_2 - x_1}{b - a}$$

$$P(42 \leq X \leq 45) = \frac{45 - 42}{47 - 41} = \frac{1}{2}$$



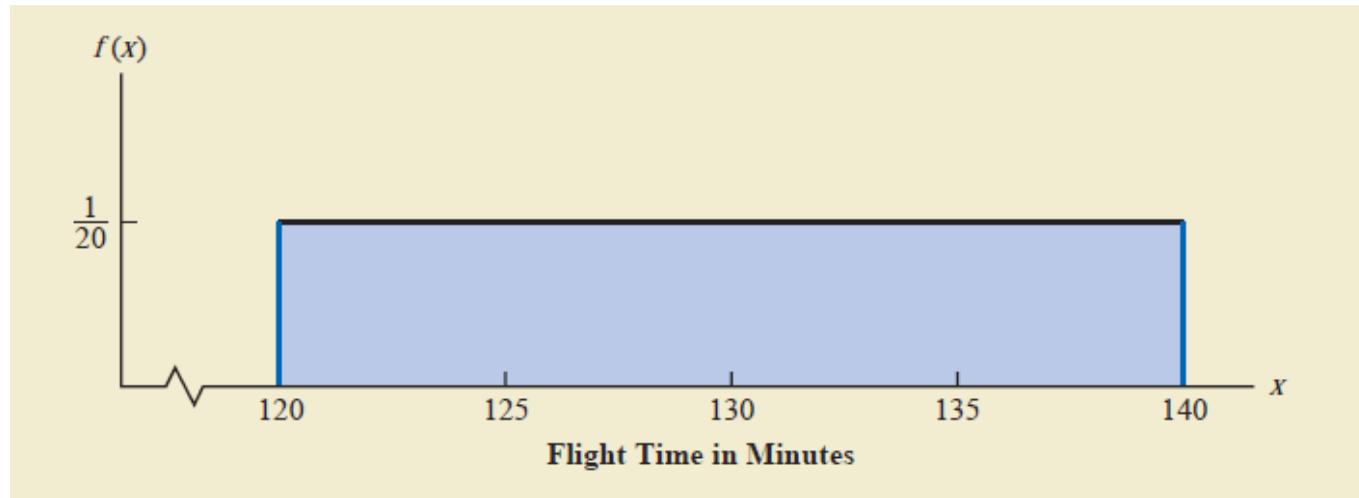
Example : Uniform Distribution

- Consider the random variable x representing the flight time of an airplane traveling from Delhi to Mumbai.
- Suppose the flight time can be any value in the interval from 120 minutes to 140 minutes.
- Because the random variable x can assume any value in that interval, x is a continuous rather than a discrete random variable

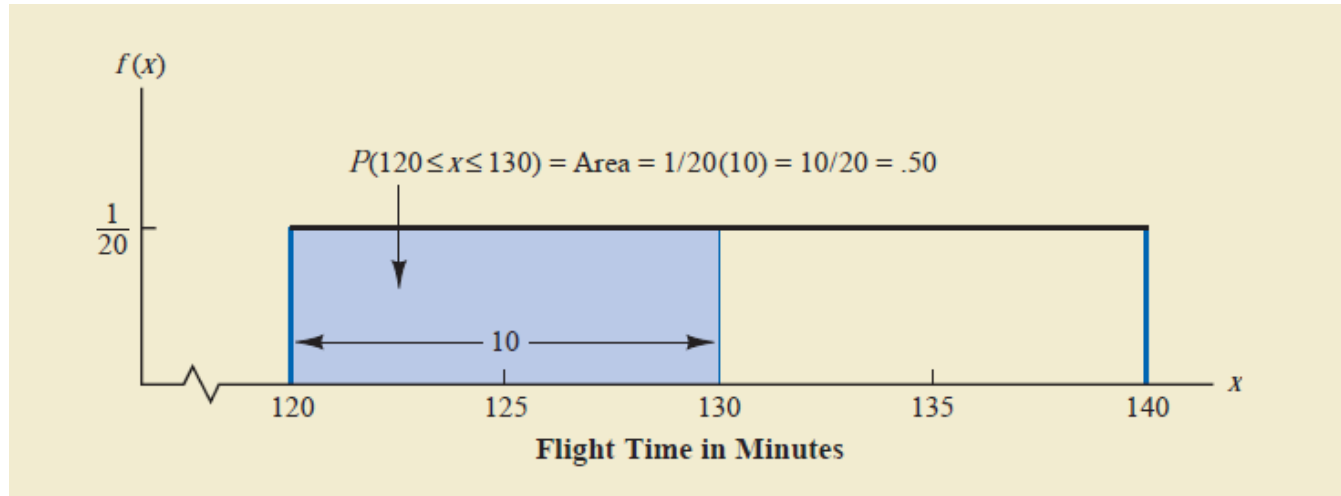
Example : Uniform Distribution contd....

- Let us assume that sufficient actual flight data are available to conclude that the probability of a flight time within any 1-minute interval is the same as the probability of a flight time within any other 1-minute interval contained in the larger interval from 120 to 140 minutes.
- With every 1-minute interval being equally likely, the random variable x is said to have a uniform probability distribution.

Uniform Probability Distribution for Flight time



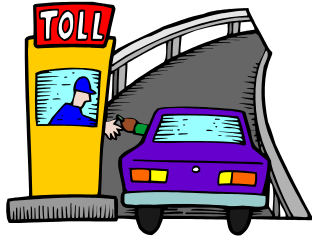
Probability of a flight time between 120 and 130 minutes



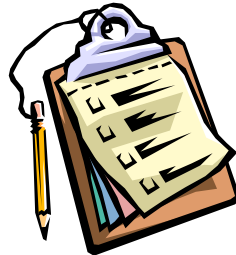
Exponential Probability Distribution

- The exponential probability distribution is useful in describing the time it takes to complete a task.
- The exponential random variables can be used to describe:

Time between
vehicle arrivals
at a toll booth



Time required
to complete
a questionnaire



Distance between
major defects
in a highway



Exponential Probability Distribution

- Density Function

$$f(x) = \frac{1}{\mu} e^{-x/\mu}$$

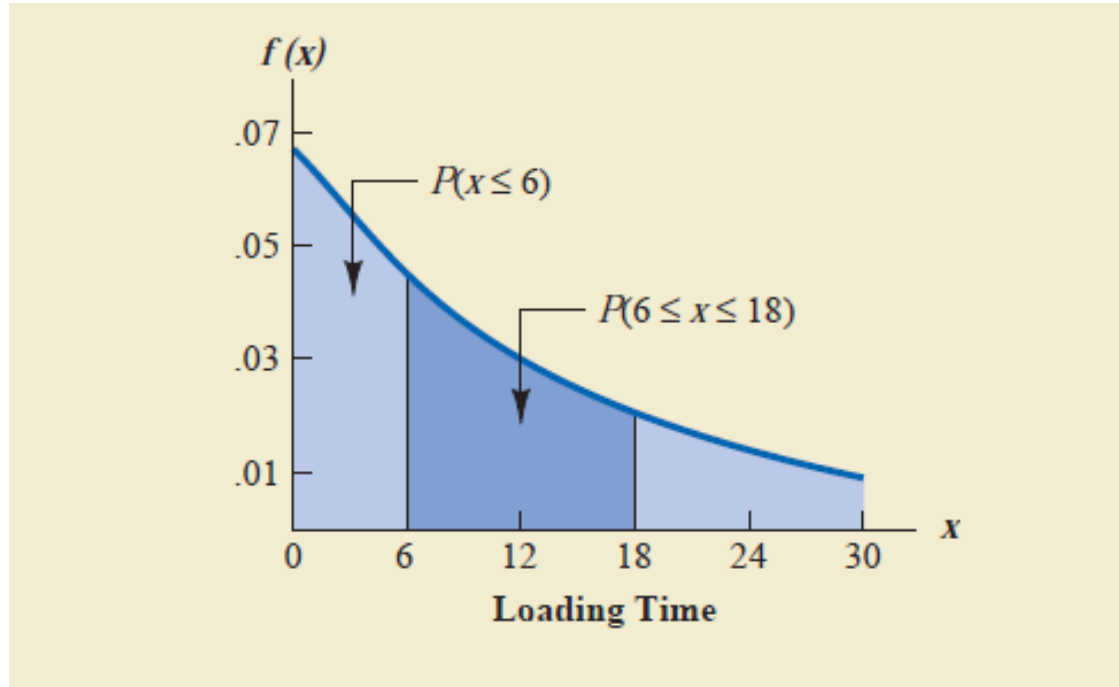
where: $\mu = \text{mean}$
 $e = 2.71828$

Exponential Probability Distribution

- Suppose that x represents the loading time for a truck at loading dock and follows such a distribution.
- If the mean, or average, loading time is 15 minutes ($\mu = 15$), the appropriate probability density function for x is

$$f(x) = \frac{1}{15} e^{-x/15}$$

Exponential Distribution for the loading Dock Example



Exponential Probability Distribution

Cumulative Probabilities

$$P(x \leq x_0) = 1 - e^{-x_0 / \mu}$$

where:

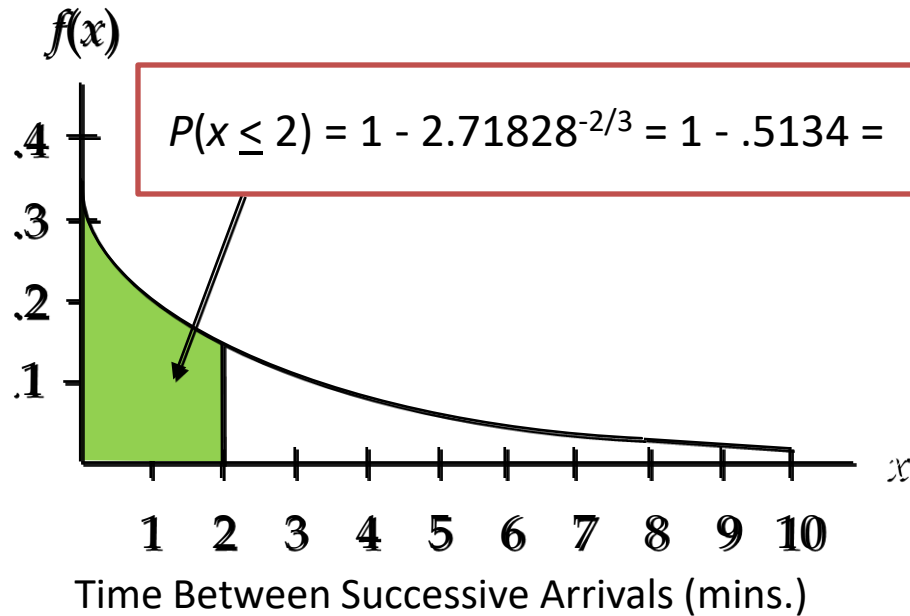
x_0 = some specific value of x

Example: Exponential Probability Distribution

- The time between arrivals of cars at a Petrol pump follows an exponential probability distribution with a mean time between arrivals of 3 minutes.
- The Petrol pump owner would like to know the probability that the time between two successive arrivals will be 2 minutes or less.

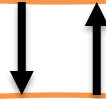


Example: Petrol Pump Problem



Relationship between the Poisson and Exponential Distributions

The Poisson distribution provides an appropriate description of the number of occurrences per interval



The exponential distribution provides an appropriate description of the length of the interval between occurrences

Mean of Poisson and Mean of Exponential Distributions

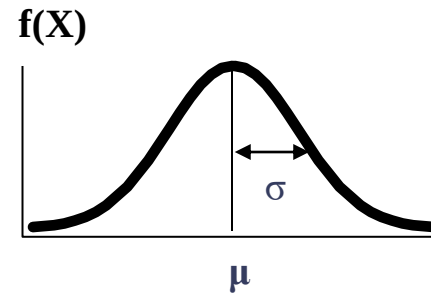
- Because the average number of arrivals is 10 cars per hour, the average time between cars arriving is

$$\frac{1 \text{ hour}}{10 \text{ cars}} = .1 \text{ hour/car}$$



The Normal Distribution: Properties

- 'Bell Shaped'
- Symmetrical
- Mean, Median and Mode are equal
- Location is characterized by the mean, μ
- Spread is characterized by the standard deviation, σ
- The random variable has an infinite theoretical range: $-\infty$ to $+\infty$



Mean = Median = Mode

The Normal Distribution: Density Function

The formula for the normal probability density function is

$$f(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2}$$

Where e = the mathematical constant approximated by 2.71828

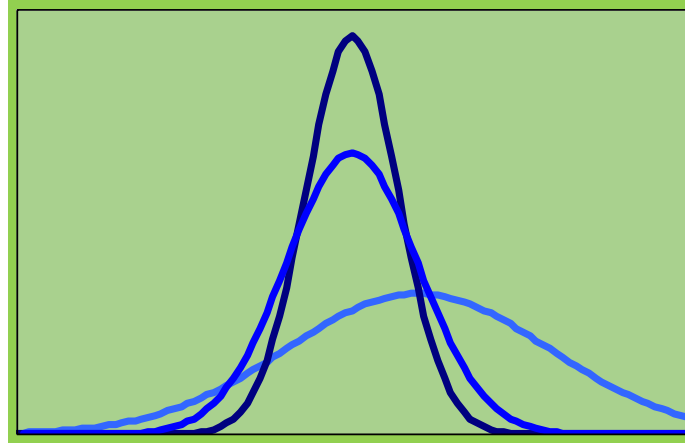
π = the mathematical constant approximated by 3.14159

μ = the population mean

σ = the population standard deviation

X = any value of the continuous variable

The Normal Distribution: Shape



By varying the parameters μ and σ , we obtain different normal distributions



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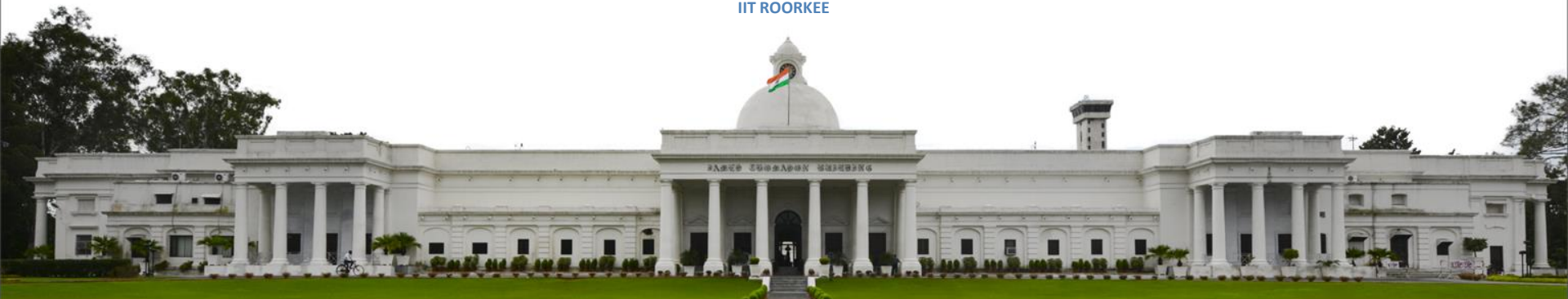


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Data Analytics with Python

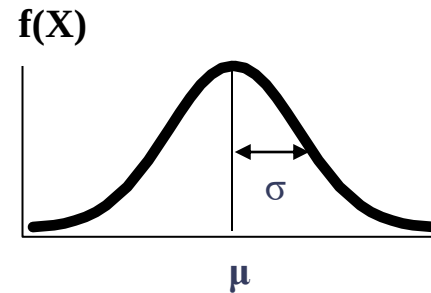
Lecture 10: Probability Distributions-III

Dr. A. Ramesh
DEPARTMENT OF MANAGEMENT
IIT ROORKEE



The Normal Distribution: Properties

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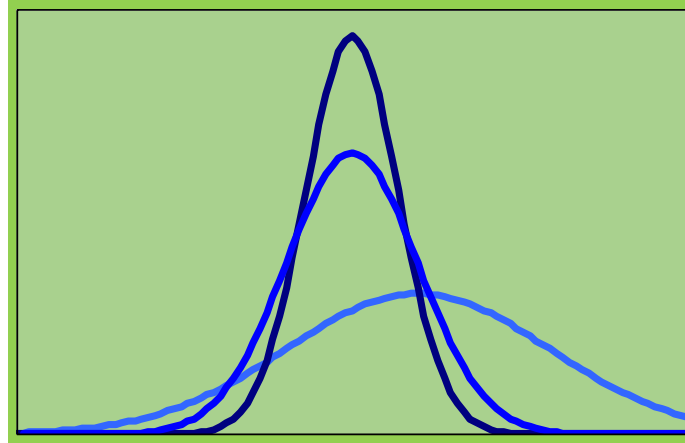
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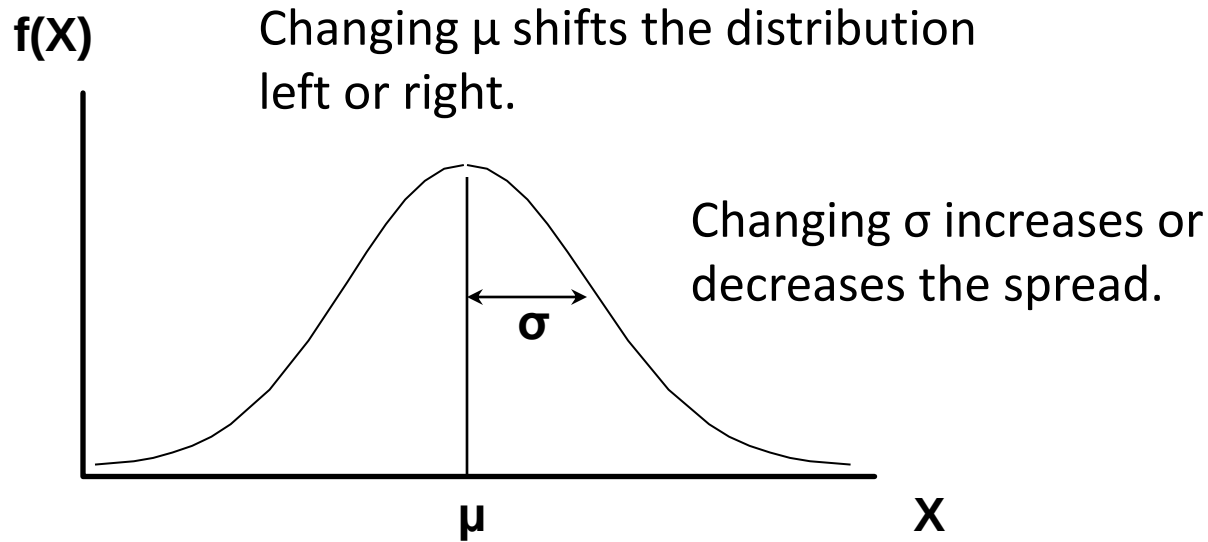
X = any value of the continuous variable

The Normal Distribution: Shape



By varying the parameters μ and σ , we obtain different normal distributions

The Normal Distribution: Shape



The Standardized Normal Distribution

- Any normal distribution (with any mean and standard deviation combination) can be transformed into the standardized normal distribution (Z).
- Need to transform X units into Z units.
- The standardized normal distribution has a mean of 0 and a standard deviation of 1.



The Standardized Normal Distribution

- Translate from X to the standardized normal (the “ Z ” distribution) by subtracting the mean of X and dividing by its standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$

The Standardized Normal Distribution: Density Function

- The formula for the standardized normal probability density function is

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}}$$

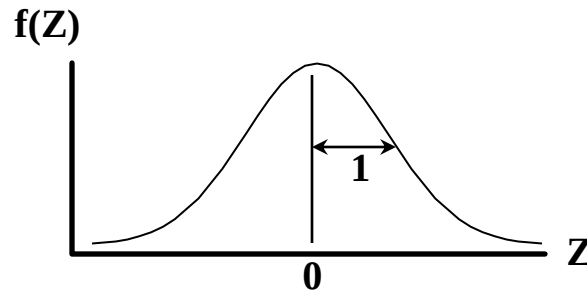
Where e = the mathematical constant approximated by 2.71828

π = the mathematical constant approximated by 3.14159

Z = any value of the standardized normal distribution

The Standardized Normal Distribution: Shape

- Also known as the “Z” distribution
- Mean is 0
- Standard Deviation is 1



Values above the mean have positive Z-values, values below the mean have negative Z-values

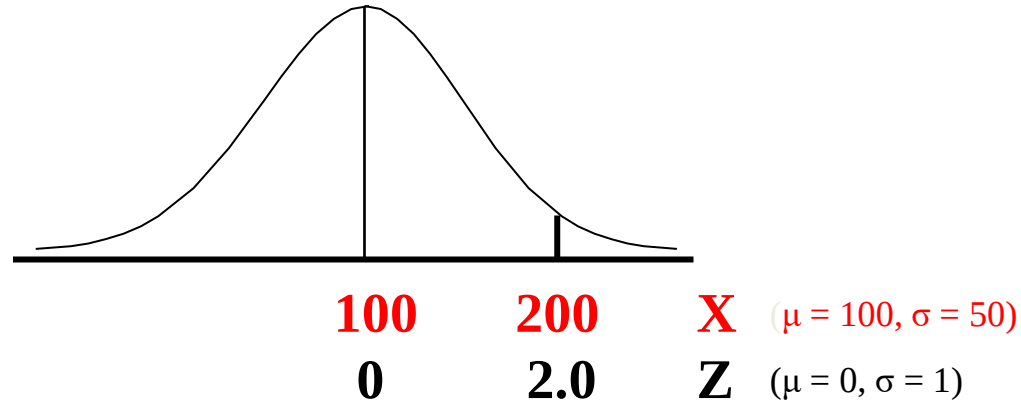
The Standardized Normal Distribution: Example

- If X is distributed normally with mean of 100 and standard deviation of 50, the Z value for $X = 200$ is

$$Z = \frac{X - \mu}{\sigma} = \frac{200 - 100}{50} = 2.0$$

- This says that $X = 200$ is two standard deviations (2 increments of 50 units) above the mean of 100.

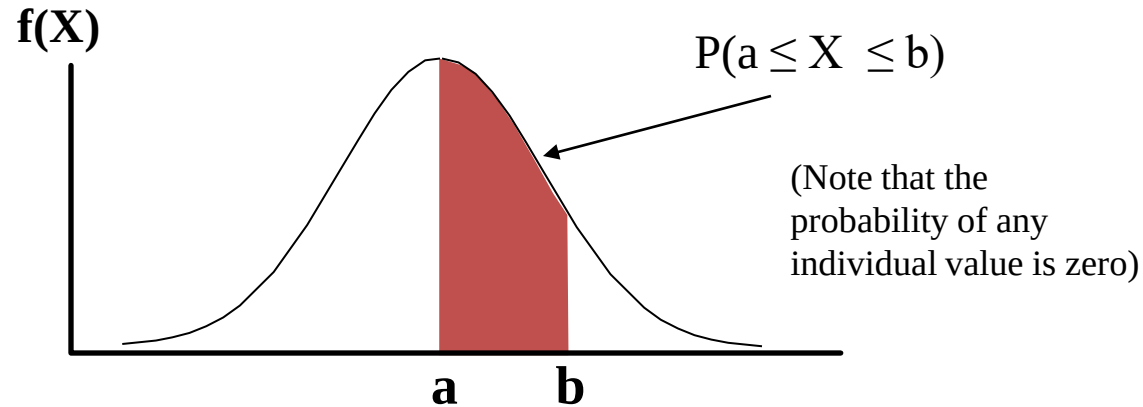
The Standardized Normal Distribution: Example



Note that the distribution is the same, only the scale has changed. We can express the problem in original units (X) or in standardized units (Z)

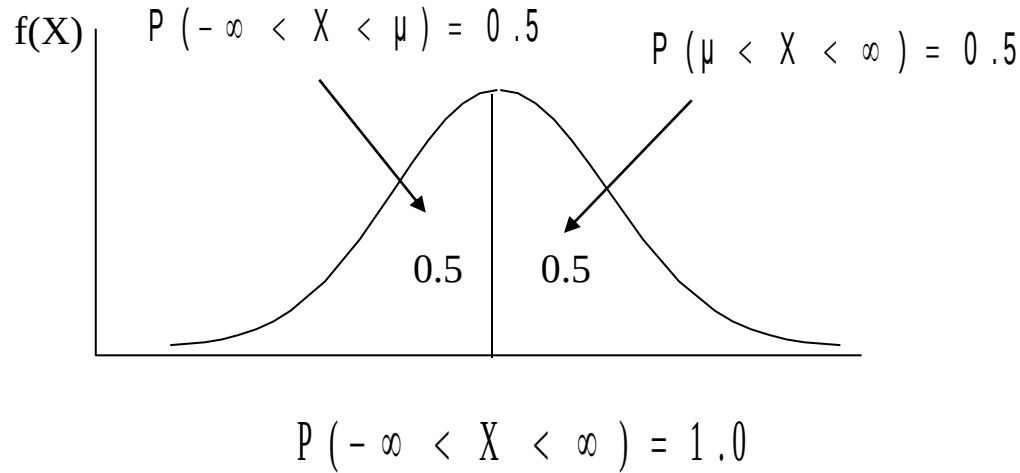
Normal Probabilities

Probability is measured by the area under the curve



Normal Probabilities

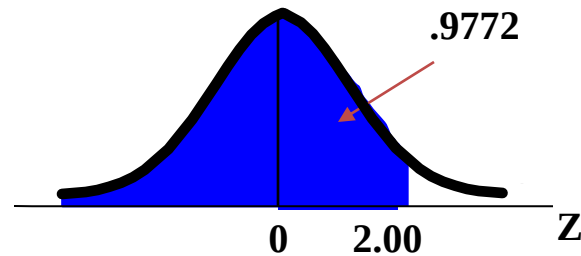
The total area under the curve is 1.0, and the curve is symmetric, so half is above the mean, half is below.



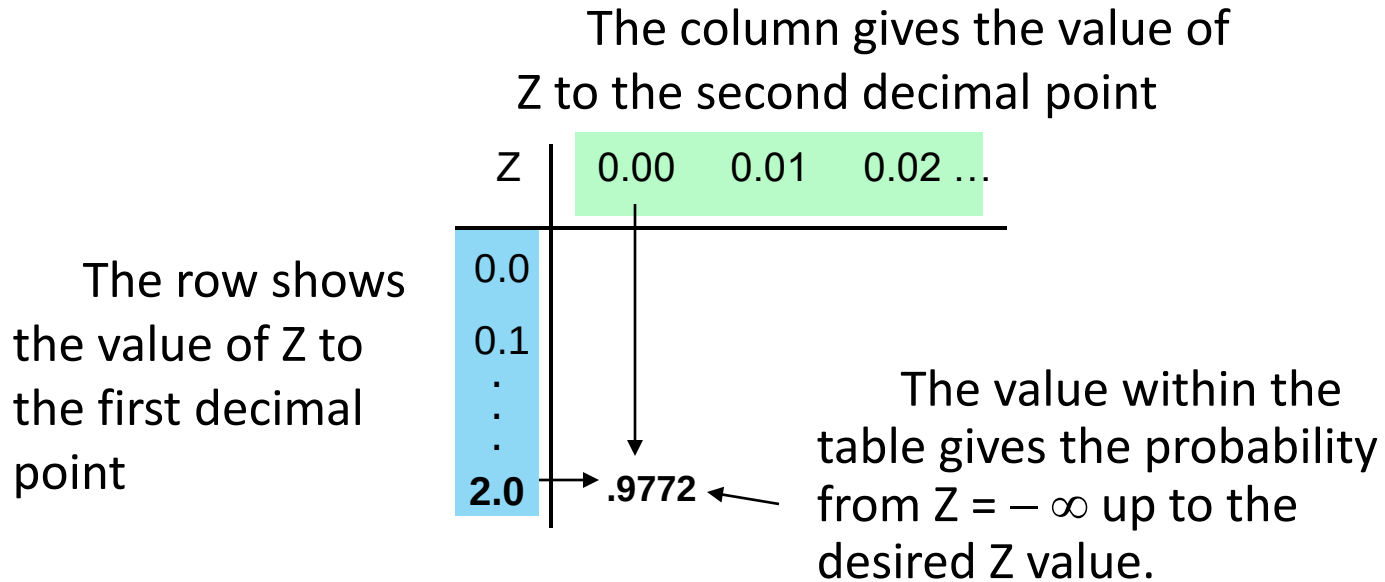
Normal Probability Tables

Example:

$$P(Z < 2.00) = .9772$$



Normal Probability Tables



$$P(Z < 2.00) = .9772$$

Finding Normal Probability Procedure

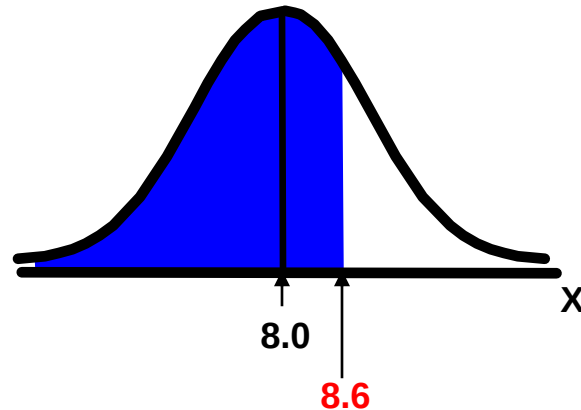
To find $P(a < X < b)$ when X is distributed normally:

- Draw the normal curve for the problem in terms of X .
- Translate X -values to Z -values.
- Use the Standardized Normal Table.



Finding Normal Probability: Example

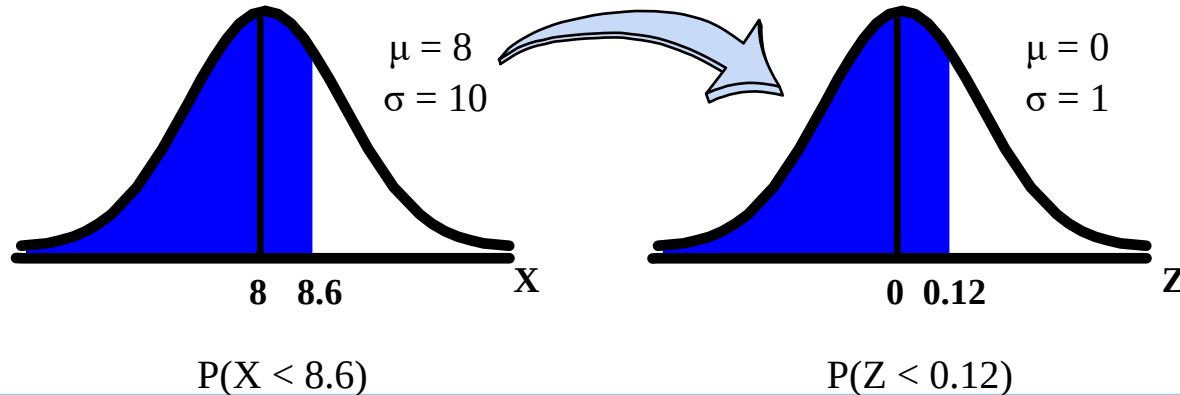
- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normal with mean 8.0 and standard deviation 5.0
- Find $P(X < 8.6)$



Finding Normal Probability: Example

- Suppose X is normal with mean 8.0 and standard deviation 5.0. Find $P(X < 8.6)$.

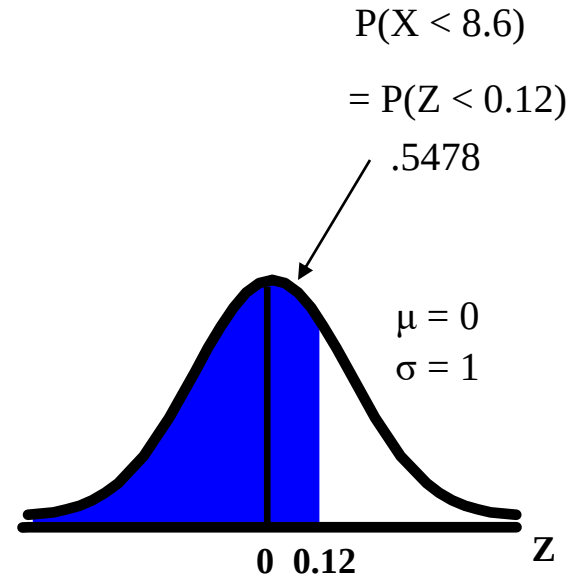
$$Z = \frac{X - \mu}{\sigma} = \frac{8.6 - 8.0}{5.0} = 0.12$$



Finding Normal Probability: Example

Standardized Normal Probability
Table (Portion)

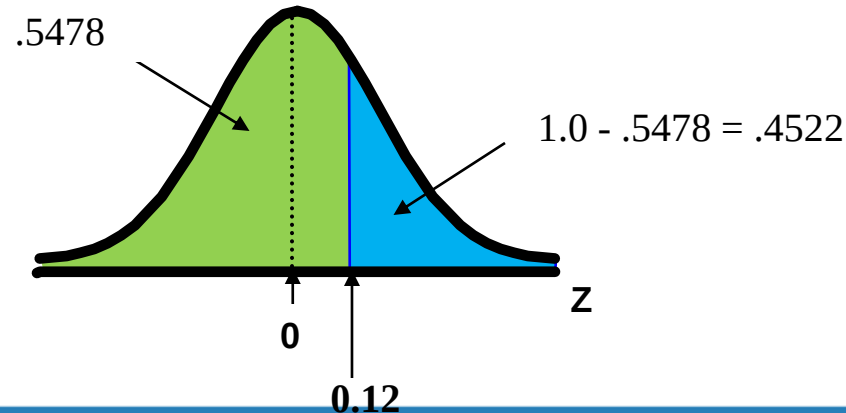
Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255



Finding Normal Probability: Example

- Find $P(X > 8.6)$...

$$\begin{aligned} P(X > 8.6) &= P(Z > 0.12) = 1.0 - P(Z \leq 0.12) \\ &= 1.0 - .5478 = .4522 \end{aligned}$$



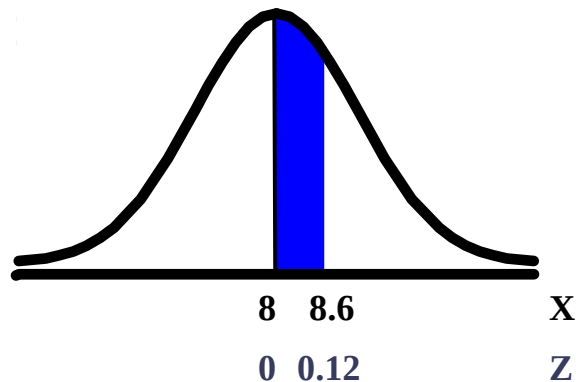
Finding Normal Probability: Between Two Values

- Suppose X is normal with mean 8.0 and standard deviation 5.0.
Find $P(8 < X < 8.6)$

Calculate Z-values:

$$Z = \frac{X - \mu}{\sigma} = \frac{8 - 8}{5} = 0$$

$$Z = \frac{X - \mu}{\sigma} = \frac{8.6 - 8}{5} = 0.12$$



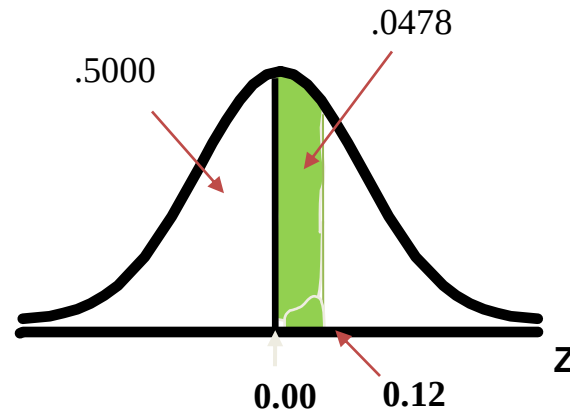
$$\begin{aligned} &P(8 < X < 8.6) \\ &= P(0 < Z < 0.12) \end{aligned}$$

Finding Normal Probability Between Two Values

- Standardized Normal Probability
- Table (Portion)

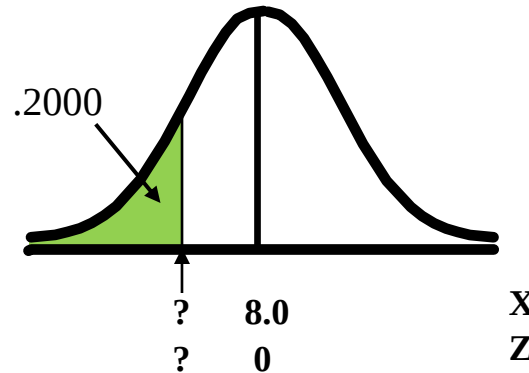
Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255

$$\begin{aligned}P(8 < X < 8.6) \\&= P(0 < Z < 0.12) \\&= P(Z < 0.12) - P(Z \leq 0) \\&= .5478 - .5000 = .0478\end{aligned}$$



Given Normal Probability: Find the X Value

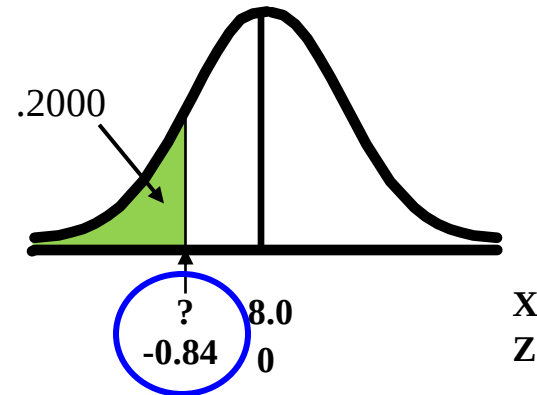
- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normal with mean 8.0 and standard deviation 5.0
- Find X such that 20% of download times are less than X .



Given Normal Probability, Find the X Value

- First, find the Z value corresponds to the known probability using the table.

Z		.03	.04	.05
-0.9		.1762	.1736	.1711
-0.8		.2033	.2005	.1977
-0.7		.2327	.2296	.2266



Given Normal Probability, Find the X Value

- Second, convert the Z value to X units using the following formula.

$$\begin{aligned}X &= \mu + Z\sigma \\&= 8.0 + (-0.84)5.0 \\&= 3.80\end{aligned}$$

So 20% of the download times from the distribution with mean 8.0 and standard deviation 5.0 are less than 3.80 seconds.

Assessing Normality

- It is important to evaluate how well the data set is approximated by a normal distribution.
- Normally distributed data should approximate the theoretical normal distribution:
 - The normal distribution is bell shaped (symmetrical) where the mean is equal to the median.
 - The empirical rule applies to the normal distribution.
 - The interquartile range of a normal distribution is 1.33 standard deviations.



Assessing Normality

- Construct charts or graphs
 - For small- or moderate-sized data sets, do stem-and-leaf display and box-and-whisker plot look symmetric?
 - For large data sets, does the histogram or polygon appear bell-shaped?
- Compute descriptive summary measures
 - Do the mean, median and mode have similar values?
 - Is the interquartile range approximately 1.33σ ?
 - Is the range approximately 6σ ?



Assessing Normality

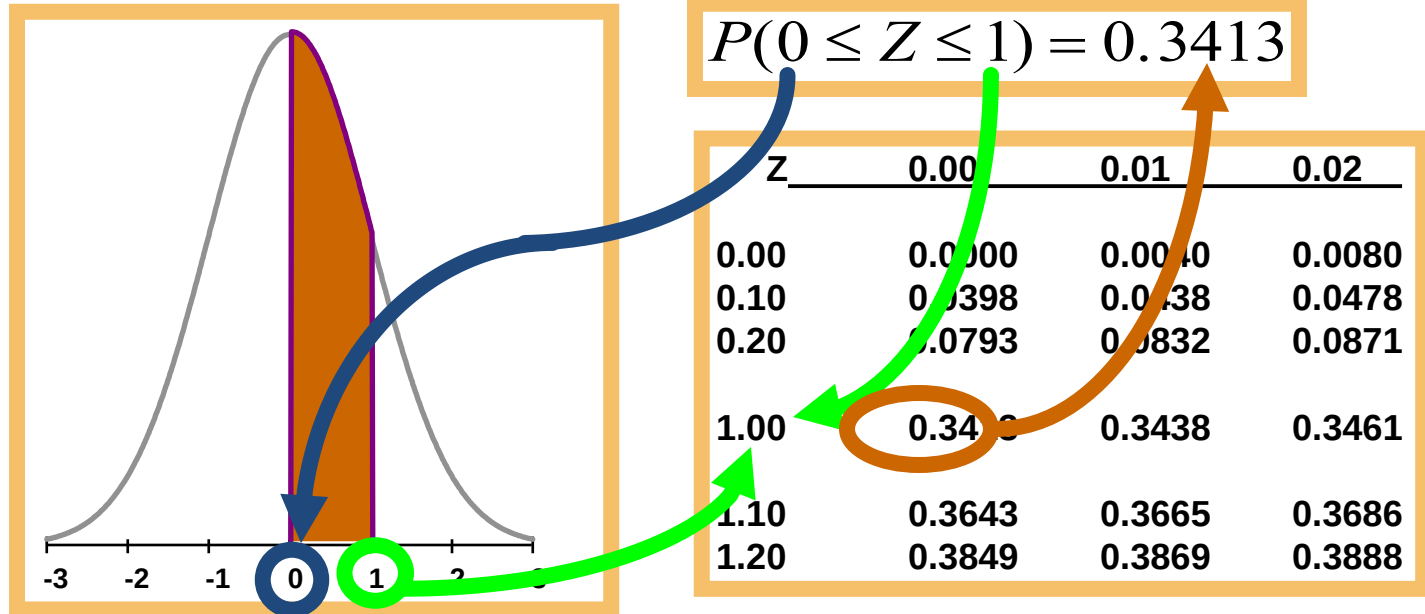
- Observe the distribution of the data set
 - Do approximately 2/3 of the observations lie within mean ± 1 standard deviation?
 - Do approximately 80% of the observations lie within mean ± 1.28 standard deviations?
 - Do approximately 95% of the observations lie within mean ± 2 standard deviations?



Z Table

Z	Second Decimal Place in Z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.10	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.20	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.30	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.90	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.00	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.10	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.20	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
2.00	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
3.00	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.40	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.50	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998

Table Lookup of a Standard Normal Probability



Applying the Z Formula

X is normally distributed with $\mu = 485$, and $\sigma = 105$

$$P(485 \leq X \leq 600) = P(0 \leq Z \leq 1.10) = .3643$$

For $X = 485$,

$$Z = \frac{X - \mu}{\sigma} = \frac{485 - 485}{105} = 0$$

For $X = 600$,

$$Z = \frac{X - \mu}{\sigma} = \frac{600 - 485}{105} = 1.10$$

Z	0.00	0.01	0.02
0.00	0.0000	0.0040	0.0080
0.10	0.0398	0.0438	0.0478
1.00	0.3413	0.3438	0.3461
1.10	0.3643	0.3665	0.3686
1.20	0.3849	0.3869	0.3888





Thank You

